

Number System

Basics of Number System

(1). **Face Value** : It is nothing but the number itself about which it has been asked.

Eg: In the number 23576, face value of 5 is 5 and face value of 7 is 7.

(2). **Place Value** : The place value of a number depends on its position in the number. Each position has a value 10^n , the places to its right.

Eg: In the number 23576, place value of 5 is 500 and place value of 3 is 3000.

Types of Numbers

(1). **Natural Numbers (N)** :

All positive counting numbers. (0 is not included in it.)

Examples: 1, 2, 3, 4 ... etc.

(2). **Whole Numbers (W)**: All non-negative numbers are all whole numbers.

Examples: 0, 1, 2, 3, 4... etc.

(3). **Integer Numbers (I)**: All positive numbers and negative numbers including zero. Positive numbers are called positive integers and negative numbers are called negative integers.

$I = \dots, -4, -3, -2, -1, 0, 1, 2, 3, 4 \dots$

(4). **Even Numbers** : 2, 4, 6, 8, 10.....

[Divisible by 2 completely]

(5). **Odd Numbers** : 1, 3, 5, 7, 9, 11..... [Not divisible by 2 completely]

(6). **Rational Numbers** : Numbers whose exact value can be determined. Also a number which can be written in the form

$\frac{p}{q}$, where p and q are integers and $q \neq 0$ is called a rational number. For example,

Examples : $\frac{3}{4} = 0.75$, $\frac{4}{5} = 0.8$, $\frac{9}{-5}$, $\frac{22}{7}$

(7). **Irrational Numbers** : Numbers whose exact value cannot be determined.

Example : $\pi = 3.142857142857 \dots$

(8). **Prime number** : A number which is divisible by 1 and itself. 2 is only an even prime number.

Example : 2, 3, 5, 7, 11, etc.

Note:-

Total prime no. between 1 - 50 $\Rightarrow 15$

Total prime no. between 1 - 100 $\Rightarrow 25$

Total prime no. between 1 - 500 $\Rightarrow 95$

Total prime no. between 1 - 1000 $\Rightarrow 168$

(9). **Composite number** : If we remove all prime numbers from natural numbers

then whatever is left is called Composite numbers.

Example : 4, 6, 8, 9, 10, 12 etc.

Note :- 1 is neither prime nor composite.

(10). **Co - prime number** : Two numbers are called Co-prime numbers if their HCF is 1.

Example : (2 and 3), (6 and 11).

Note : Two prime numbers are always co-prime numbers to each other. Any two consecutive integers are always co-prime numbers to each other.

Factors

The factors of a number are the numbers that divide it completely without leaving any remainder.

Example : 24 can be completely divided by 1, 2, 3, 4, 6, 8, 12 and 24, so these numbers are factors of 24.

Prime factorisation of a number : When a number is written in the form of multiplication of its prime factors, it's called prime factorisation.

Prime factorisation of 24.

2	24
2	12
2	6
2	3

$$24 \rightarrow 2 \times 2 \times 2 \times 3 \text{ or } 2^3 \times 3^1$$

Number of factors : To find the number of factors we write the number in the form of prime factors and then add +1 to the exponent of prime factors and multiply them.

For example: $24 = 2^3 \times 3^1$

Number of factors of 24 $\rightarrow (3+1)(1+1) = 4 \times 2 = 8$.

With the help of an example, we try to find the sum of all factors of a number.

$$24 = 2^3 \times 3^1,$$

$$\text{Sum of all factors} = (2^0 + 2^1 + 2^2 + 2^3) \times (3^0 + 3^1) = 15 \times 4 = 60.$$

Number of even factors of a number : To find the number of even factors of a number, we add +1 to the exponents of prime numbers except 2.

(Note : If a number doesn't have 2 as its factor it will have 0 even factors)

Que. Find the number of even factors of 120.

$$\text{Ans. } 120 = 2^3 \times 3^1 \times 5^1$$

Number of even factors = $3 \times (1+1) \times (1+1) = 3 \times 2 \times 2 = 12$

Note :- To find the sum of even factors, we shall ignore 2^0 ,

Que. Find the sum of even factors of 120.

$$\text{Sol:- Sum of even factors} = (2^1 + 2^2 + 2^3)(3^0 + 3^1)(5^0 + 5^1) = 14 \times 4 \times 6 = 336.$$

Number and Sum of odd factors of a number : to find the number and sum of odd factors of a number, we have to ignore the exponents of 2.

Que. Find the number of odd factors 120.

$$\text{Sol:- } 120 = 2^3 \times 3^1 \times 5^1$$

$$\text{Required number} = (1+1)(1+1) = 4$$

The exponent of 2 is completely ignored.

$$\text{Sum of odd factors of } 120 = (3^0 + 3^1)(5^0 + 5^1) = 4 \times 6 = 24$$

Some Important Results of Factors:

$$1001 = 7 \times 11 \times 13$$

$$1001 \times abc = abcabc$$

$$1001 \times 234 = 234234$$

Que: Which of the following is a factor of 531531?

(a) 15 (b) 13 (c) 11 (d) both b and c

$$\text{Sol:- } 531531 = 1001 \times 531$$

$= 7 \times 11 \times 13 \times 531$ So, both 11 and 13 are factors of 531531.

$$111 = 37 \times 3, 1001 \times 111 = 111111,$$

When a single digit is written 6 times, 3, 7, 11, 13, and 37 are factors of it.

Que. Which of the following is a factor of 222222?

(a) 17 (b) 57 (c) 68 (d) 74

$$\text{Sol:- } 222222 = 2 \times 111111$$

$$= 2 \times 3 \times 7 \times 11 \times 13 \times 37$$

Clearly, $2 \times 37 = 74$ is one of the factors.

$\rightarrow$ If a, b and c are prime numbers, then the number of prime factors of $a^x \times b^y \times c^z$ is $(x+y+z)$.

Recurring Decimal

Recurring decimals are referred to as numbers that are uniformly repeated after the decimal. Some rational numbers produce recurring decimals after converting them into decimal numbers, but all irrational numbers produce recurring decimals after converting them into decimal form.

Examples :

$$(1) \frac{1}{3} = 0.3333333 \dots = 0.\bar{3}$$

$$(2) 0.\bar{9} = \frac{9}{9} = 1$$

$$(3) 0.53\bar{27} = \frac{5327 - 53}{9900} = \frac{5274}{9900}$$

$$(4) 2.53\bar{27} = 2 + \frac{5327 - 53}{9900} = 2\frac{5274}{9900}$$

Divisibility Test

By 2:- When last digit is 0 or an even

number eg: 520, 588

By 3:- Sum of digits is divisible by 3
eg: 1971, 1974

By 4:- When last two digits are divisible by 4 or, they are zeros eg: 1528, 1700

By 5 :- When last digit is 0 or 5
eg: 1725, 1790

By 6 :- When the number is divisible by 2 and 3 both. eg: 36, 72

By 7 :- Subtract twice the last digit from the number formed by the remaining digits. Like 651 divisible by 7
 $65 - (1 \times 2) = 63$. Since 63 is divisible by 7, so is 651.

By 8 :- When the last three digits are divisible by 8. eg: 2256

By 9 :- When sum of digit is divisible by 9
eg: 9216

By 10 :- When the last digit is 0. eg: 452600

By 11:- When the difference between the sum of odd and even place digits is equal to 0 or multiple of 11.
eg: 217382

Sum of odd place digits = $2 + 7 + 8 = 17$
Sum of even place digits = $1 + 3 + 2 = 6$
 $17 - 6 = 11$, hence 217382 is divisible by 11.

By 13 :- If adding four times the last digit to the number formed by the remaining digits is divisible by 13, then the number is divisible by 13. Like 1326 is divisible by 13

$132 + (6 \times 4) = 156$. Repeat the same process for 156.

$15 + (6 \times 4) = 39$. so 39 is divisible by 13

BY 17 :- The divisibility rule of 17 states, "If five times the last digit is subtracted from a number made up of the remaining digits and the remainder is either 0 or a multiple of 17, then the number is divisible by 17".

Like 221: $22 - 1 \times 5 = 17$.

Prime Number Test

For finding whether any number is a prime number or not, we need to find the nearest square root of given number, then we need to find out whether the given number is divisible by any prime number less than the obtained number or not. If it is divisible then it is not a prime number and if not divisible then it is a prime number.

Example : Find whether 177 is a prime

number or not.

Soln : Nearest square root of 177 is 13. Now we need to check whether 177 is divisible by prime numbers less than 13. On checking we find that 177 is divisible by 3. Hence, 177 is not a prime number.

Important Formulas

1. Sum of first n natural number

$$s = \frac{n(n+1)}{2}$$

2. Sum of first n odd numbers = n^2

3. Sum of first n even numbers
= $n(n+1)$

4. Sum of square of first n natural numbers = $\frac{n(n+1)(2n+1)}{6}$

5. Sum of cubes of first n natural number = $\left(\frac{n(n+1)}{2}\right)^2$

6. $(x^m - a^m)$ is divisible by $(x - a)$ for every natural number m .

7. $(x^m - a^m)$ is divisible by $(x + a)$ and $(x - a)$ for even values of m .

8. $(x^m + a^m)$ is divisible by $(x + a)$ for odd values of m .

9. Number of prime factors of a^p, b^q, c^r, d^s is $p + q + r + s$ when a, b, c, d are all prime numbers.

10. HCF of $(a^m - 1)$ and $(a^n - 1) = [(a^{\text{HCF}(m,n)} - 1)]$

Number of Zeros in an expression

We shall understand this concept with the help of an example.

Let's find the number of zeros in the following expression: $24 \times 32 \times 17 \times 23 \times 19 = (2^3 \times 3^1) \times 2^5 \times 17 \times 23 \times 19$
Notice that a zero is made only when there is a combination of 2 and 5. Since there is no '5' here there will be no zero in the above expression.

Example:-

$8 \times 15 \times 23 \times 17 \times 25 \times 22 = 2^3 \times (3^1 \times 5^1) \times 23 \times 17 \times 5^2 \times 2^1 \times 11$
In this expression there are 4 twos and 3 fives. From this 3 pairs of 5×2 can be formed. Therefore, there will be 3 zeros in the final product.

Que. Find the number of zeros in the value of:

$$2^2 \times 5^4 \times 4^6 \times 10^8 \times 6^{10} \times 15^{12} \times 8^{14} \times 20^{16} \times 10^{18} \times 25^{20}$$

Sol:-

$$2^2 \times 5^4 \times 4^6 \times 10^8 \times 6^{10} \times 15^{12} \times 8^{14} \times 20^{16} \times 10^{18} \times 25^{20} = 2^2 \times 5^4 \times 2^{12} \times 2^8 \times 5^8 \times 2^{10} \times 3^{10} \times 3^{12} \times 5^{12} \times 2^{42} \times 2^{32} \times 5^{16} \times 2^{18} \times 5^{18} \times 5^{40}$$

Zeros are possible with a combination of 2×5 . Here the number of 5's are less so the number of zeros will be limited to the number of 5's.
In this expression number of fives are:
 $5^4 \times 5^8 \times 5^{12} \times 5^{16} \times 5^{18} \times 5^{40}$;
i.e. $4 + 8 + 12 + 16 + 18 + 40 = 98$

The number of Zeros in $n!$

To find the number of zeros in $n!$, we divide "n" by 5 until we get a number less than 5, and then we add all the quotients so obtained.

Que. Find the number of zeros in $36!$.

5	36
5	7 (1)
	1 (2)

The number of zeros = $7 + 1 = 8$.

Remainder Theorem

Que. What will be the remainder when 17×23 is divided by 12?

Ans :- We can express this as:

$$17 \times 23 = (12 + 5) \times (12 + 11) = 12 \times 12 + 12 \times 11 + 5 \times 12 + 5 \times 11$$

In the above expression we will find that remainder will depend on the last term i.e. 5×11

$$\text{Now, } \text{rem.} \left(\frac{5 \times 11}{12} \right) = 7.$$

$$\text{So, } \frac{12 \times 12 + 12 \times 11 + 5 \times 12 + 5 \times 11}{12}$$

and $\frac{5 \times 11}{12}$ remainder is same in both cases which is 7.

Example:- Remainder when

$1421 \times 1423 \times 1425$ is divided by 12?

$$\begin{aligned} \text{rem.} \left(\frac{1421 \times 1423 \times 1425}{12} \right) \\ = \text{rem.} \left(\frac{5 \times 7 \times 9}{12} \right) &= \text{rem.} \left(\frac{35 \times 9}{12} \right) \\ = \text{rem.} \left(\frac{11 \times 9}{12} \right) &= 3 \end{aligned}$$

Negative Remainder

Taking a negative remainder will make our calculation easier.

Examples

$$(i) \text{ rem.} \left(\frac{7 \times 8}{9} \right) = \text{rem.} \left(\frac{-2 \times -1}{9} \right)$$

$$= -2 \times -1 = 2$$

$$(ii) \text{ rem.} \left(\frac{55 \times 56}{57} \right) = \text{rem.} \left(\frac{-2 \times -1}{57} \right)$$

$$= -2 \times -1 = 2$$

$$(iii) \text{ rem } \left(\frac{7 \times 10}{9} \right) = \text{rem} \left(\frac{-2 \times 1}{9} \right)$$

$$= -2 \times 1 = -2 \text{ or, } 7$$

Large Power Concepts

Look at the following examples:

$$(i) \text{ rem } \left(\frac{28^{12345}}{9} \right) = \text{rem} \left(\frac{(27+1)^{12345}}{9} \right)$$

$$= \text{rem} \left(\frac{1^{12345}}{9} \right) = 1^{12345} = 1$$

$$(ii) \text{ rem } \left(\frac{26^{12345}}{9} \right)$$

$$= \text{rem} \left(\frac{(27-1)^{12345}}{9} \right)$$

$$= \text{rem} \left(\frac{-1^{12345}}{9} \right) = -1^{12345} = -1 \text{ or } 8$$

Application of Remainder Theorem

Que. Find the last two digits of the expression

$$22 \times 31 \times 44 \times 27 \times 37 \times 43?$$

Sol:- If we divide the above expression by 100, we will get the last two digits as remainder.

$$\Rightarrow \text{rem} \left(\frac{22 \times 31 \times 44 \times 27 \times 37 \times 43}{100} \right)$$

dividing by 4 to make it simple

$$= \text{rem} \left(\frac{22 \times 31 \times 11 \times 27 \times 37 \times 43}{25} \right)$$

$$= \text{rem} \left(\frac{132 \times 22 \times 216}{25} \right)$$

$$= \text{rem} \left(\frac{7 \times 22 \times 16}{25} \right) \Rightarrow \text{rem} \left(\frac{4 \times 16}{25} \right)$$

$$= \text{rem} \left(\frac{14}{25} \right) = 14$$

Since we had divided by 4 initially now to get the correct answer, we need to multiply the remainder by 4.

So remainder will be $14 \times 4 = 56$, which will also be the last two digits of the expression.

Variety Questions

Q.1. Find the value of x in given equation:

$$\frac{\sqrt{5+x} + \sqrt{5-x}}{\sqrt{5+x} - \sqrt{5-x}} = 7$$

SSC CGL 20/09/2025 (Shift 1)

$$(a) \frac{12}{5} \quad (b) \frac{7}{5} \quad (c) \frac{6}{5} \quad (d) \frac{8}{5}$$

Q.2. A gardener organizes trees into rows to create a square formation but discovers that 16 trees are not included. If the total number of trees is 4112, then what is the number of trees in each row?

SSC CGL 17/09/2025 (Shift 3)

$$(a) 42 \quad (b) 54 \quad (c) 64 \quad (d) 65$$

Q.3. The difference between the two

numbers is 1370. On dividing the larger by the smaller, the quotient is 7, and the remainder is 14. Find the smaller number.

Graduate Level 28/07/2025 (Shift - 3)

$$(a) 240 \quad (b) 226 \quad (c) 285 \quad (d) 295$$

Q.4. Converting the decimal number 187 to the binary number system we get:

Higher Secondary 29/07/2025 (Shift - 2)

$$(a) 10111011 \quad (b) 10111010$$

$$(c) 11011101 \quad (d) 11011110$$

Q.5. Which set of numbers includes all the other sets?

Higher Secondary 28/07/2025 (Shift - 2)

$$(a) \text{Integers} \quad (b) \text{Rational numbers}$$

$$(c) \text{Real numbers} \quad (d) \text{Natural numbers}$$

Q.6. How many non-square natural numbers lie between 30^2 and 31^2 ?

Matriculation Level 29/08/2025 (Shift - 3)

$$(a) 59 \quad (b) 60 \quad (c) 61 \quad (d) 62$$

Q.7. What will be the number of zeros at the end of the square of the following numbers?

$$(i) 90$$

$$(ii) 600$$

Choose the correct option:

Matriculation Level 02/08/2025 (Shift - 1)

$$(a) 1 \text{ zero in (i), 3 zeros in (ii)}$$

$$(b) 2 \text{ zeros in (i), 4 zeros in (ii)}$$

$$(c) 2 \text{ zeros in (i), 2 zeros in (ii)}$$

$$(d) 1 \text{ zero in both (i) and (ii)}$$

Q.8. If $x = \frac{5}{3}$ is a rational number and $y =$

$\sqrt{5}$, an irrational number, then $x + y$ is:

Matriculation Level 02/08/2025 (Shift - 1)

$$(a) \text{Rational} \quad (b) \text{Irrational}$$

$$(c) \text{Integer} \quad (d) \text{Whole Number}$$

Q.9. The five - digit number 45yz0 is divisible by 40. What is the maximum possible value of $(y + z)$?

$$(a) 15 \quad (b) 18 \quad (c) 16 \quad (d) 17$$

Q.10. In a room, there are some chairs and some people. If on each chair, only one person is seated, then there is no chair for exactly one person. If on each chair, two persons sit, then there is one vacant chair. What is the number of chairs in the room?

$$(a) 7 \quad (b) 4 \quad (c) 3 \quad (d) 6$$

Q.11. Cindy bought 15 apples and 12 oranges and paid a sum of ₹447 for the purchase. Which of the statements given below is inconsistent with the information given in the previous statement, leading to no possible prices of the two fruits?

$$(a) \text{Purchased 25 apples and 20 oranges}$$

and paid a sum of ₹745

$$(b) \text{Purchased 10 apples and 13 oranges and paid a sum of ₹353}$$

$$(c) \text{Purchased 12 apples and 8 oranges and paid a sum of ₹340}$$

$$(d) \text{Purchased 35 apples and 28 oranges and paid a sum of ₹1,029}$$

Q.12. Aman has certain number of notes of denomination ₹20 and ₹10 which amount to ₹390. If the numbers of notes of each kind are interchanged, then the new amount is ₹90 less than before. Find the number of notes of ₹20 denomination.

$$(a) 16 \quad (b) 12 \quad (c) 15 \quad (d) 14$$

Q.13. 2 biscuits and 1 chocolate cost ₹69. 2 chocolates and 3 cups of coffee cost ₹127. 3 biscuits, 4 chocolates and 2 cups of coffee cost ₹229. Find the total cost (in ₹) of 5 biscuits, 5 chocolates and 5 cups of coffee.

$$(a) 304 \quad (b) 375 \quad (c) 345 \quad (d) 355$$

Q.14. 8 apples and 10 oranges weigh 5 kg. 12 apples and 20 oranges weigh 9 kg. What is the weight (in kg, rounded off to the nearest integer) of 15 apples and 24 oranges?

$$(a) 16 \quad (b) 11 \quad (c) 13 \quad (d) 10$$

Q.15. In an election, voters have the option to vote for one of three candidates. They also have the option to select 'None Of The Above' (NOTA) to reject all the candidates. All three candidates get 6 times the NOTA votes. The winner gets twice the votes of the second runner-up. The first runner-up gets 900 more votes than NOTA and defeats the second runner-up by 150 votes. Find the total number of votes cast.

$$(a) 15177 \quad (b) 11577 \quad (c) 11025 \quad (d) 17715$$

Q.16. How many factors of 14,400 are divisible by 18 but not by 36?

$$(a) 3 \quad (b) 5 \quad (c) 4 \quad (d) 2$$

Q.17. Raju has ₹11,000 and starts saving ₹5,000 each week towards buying a new laptop. At the same time, Ramesh has ₹60,000 and begins spending ₹2,000 per week on supplies for his art class. Will there be a week when they have the same amount of money?

$$(a) \text{Yes, after 5 weeks}$$

$$(b) \text{No, they will never have the same amount}$$

$$(c) \text{Yes, after 7 weeks}$$

$$(d) \text{Yes, after 6 weeks}$$

Q.18. If $3c2933k$ is divisible by both 5 and 11, where c and k are single digit natural numbers, then $c + k =$ _____.



(a) 6 (b) 8 (c) 5 (d) 7

Q.19. If two numbers are each divided by the same divisor, then the remainders are 6 and 7, respectively. If the sum of the two numbers be divided by the same divisor, then the remainder is 5. The divisor is:

(a) 6 (b) 4 (c) 13 (d) 8

Q.20. Find the lowest positive value of (c - b) such that the 7-digit number 1738b9c is divisible by 12.

(a) 4 (b) 7 (c) 1 (d) 2

Q.21. Successive division of a number by 2, 3, 5 and 7 gives remainders 1, 4, 0 and 5, respectively. What will be the sum of the remainders if the same number is divided by 7, 5, 3 and 2 successively?

(a) 17 (b) 8 (c) 10 (d) 9

Q.22. Ram gives a six-digit number 468312 to Shyam to check the divisibility. Shyam tells Ram that the number is divisible by 57. Shyam asks Ram, "If we rearrange the digits of this number in descending order, then by which number will it be always divisible?"

(a) 2 (b) 3 (c) 19 (d) 17

Q.23. A 4-digit number '34PQ' divisible by 3, 5 and 7. Find the value of P + Q.

(a) 11 (b) 12 (c) 10 (d) 13

Q.24. A six-digit number is divisible by 33. If 21 is added to the number, then the new number which formed is also divisible by:

(a) 5 (b) 4 (c) 2 (d) 3

Q.25. A six-digit number 11p9q4 is divisible by 24. Then the greatest possible value for pq is:

(a) 56 (b) 68 (c) 42 (d) 32

Q.26. The remainder of the term $9 + 9^2 + \dots + 9^{(2n+1)}$ when divided by 6 is:

(a) 1 (b) 4 (c) 2 (d) 3

Q.27. Two numbers, when divided by a certain divisor, leave the remainder 57. When sum of the two numbers is divided by the same divisor, the remainder is 49. The divisor is:

(a) 56 (b) 57 (c) 49 (d) 65

Q.28. In a division sum, the divisor is 11 times the quotient and 5 times the remainder. If the remainder is 44, then the dividend is:

(a) 8888 (b) 4448 (c) 8444 (d) 4444

Q.29. What is the least value of x + y, if 10 digit number 780x533y24 is divisible by 88?

(a) 4 (b) 3 (c) 1 (d) 2

Q.30. Suhas mistakenly took as dividend a number which was 10% less than the original dividend. He also mistakenly took as divisor a number which was 20% less than the original divisor. If the correct quotient of the original question of division was 24 and the remainder was 0, then what quotient did Suhas obtain, assuming there was no error in his calculations?

(a) 27 (b) 21.6 (c) 26.4 (d) 30

Q.31. During a division, Pranjal mistakenly took as the dividend a number that was 10% more than the original dividend. He also mistakenly took as the divisor a number that was 25% more than the original divisor. If the correct quotient of the original division problem was 25 and the remainder was 0, what was the quotient that Pranjal obtained, assuming his calculations had no error?

(a) 21.75 (b) 21.25 (c) 28.75 (d) 22

Q.32. The six-digit number $7x1yyx$ is a multiple of 33 for non-zero digits x and y. Which of the following could be a possible value of (x + y)?

(a) 5 (b) 4 (c) 2 (d) 3

Q.33. A girl wants to plant trees in her garden in rows in such a way that the number of trees in each row to be the same. There are 10 rows and the number of trees in each row is 12, what is the number of trees in each row, if there are 5 more rows?

(a) 10 (b) 8 (c) 6 (d) 12

Q.34. What is the total number of factors of the number 720 except 1 and the number itself?

(a) 29 (b) 27 (c) 32 (d) 28

Q.35. Which of the following is the smallest among $(14)^{\frac{1}{3}}$, $(12)^{\frac{1}{2}}$, $(16)^{\frac{1}{6}}$ & $(25)^{\frac{1}{12}}$?

(a) $(14)^{\frac{1}{3}}$ (b) $(25)^{\frac{1}{12}}$ (c) $(16)^{\frac{1}{6}}$ (d) $(12)^{\frac{1}{2}}$

Q.36. How many terms of the series - 9, - 6, - 3, must be taken so that the sum of all the terms is 45?

(a) 11 (b) 8 (c) 10 (d) 9

Q.37. Which of the following statements is correct?

I. The Value of $100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + 94^2 - 93^2 + \dots + 22^2 - 21^2$ is 4840.

II. The value of $(K^2 + \frac{1}{K^2})(K - \frac{1}{K})(K^4$

 $+ \frac{1}{K^4})(K + \frac{1}{K})(K^4 - \frac{1}{K^4})$ is $K^{16} - \frac{1}{K^{16}}$

(a) Neither I nor II (b) Both I and II
(c) Only II (d) Only I

Q.38. If the seven-digit number 52A6B7C is divisible by 33, and A, B, C are primes, then the maximum value of $2A + 3B + C$ is:

(a) 32 (b) 23 (c) 27 (d) 34

Q.39. In a test (+ 5) marks are given for every correct answer and (-2) marks are given for every incorrect answer. Jay answered all the questions and scored (-12) marks, though he got 4 correct answers. How many of his answers were INCORRECT?

(a) 8 (b) 32 (c) 16 (d) 20

Q.40. The number 150328 is divisible by 23. If the digits are rearranged in descending order and five times of 13 is subtracted from the new number thus formed, then the resultant number will be divisible by:

(a) 3 (b) 5 (c) 11 (d) 2

Q.41. What is the sum of all the common terms between the given series S1 and S2?

S1 = 2, 9, 16,, 632

S2 = 7, 11, 15,, 743

(a) 6974 (b) 6750 (c) 7140 (d) 6860

Q.42. Let p, q, r and s be positive natural numbers having three exact factors including 1 and the number itself. If $q > p$ and both are two-digit numbers, and $r > s$ and both are one-digit numbers, then the value of the expression $\frac{p-q-1}{r-s}$ is:

(a) -s-1 (b) s-1 (c) 1-s (d) s+1

Q.43. Three fractions x, y and z are such that $x > y > z$. When the smallest of them is divided by the greatest, the result is $\frac{9}{16}$ which exceeds y by 0.0625. If $x + y + z = 2 - \frac{3}{12}$, then what is the value of x + z?

(a) $\frac{5}{4}$ (b) $\frac{1}{4}$ (c) $\frac{7}{4}$ (d) $\frac{3}{4}$

Q.44. Find the number of prime factors in the product $(30)^5 \times (24)^5$.

(a) 45 (b) 35 (c) 10 (d) 30

Q.45. Let ab, a ≠ b, is a 2-digit prime number such that ba is also a prime number. The sum of all such number is:

(a) 374 (b) 418 (c) 407 (d) 396

Q.46. Given that $2^{20} + 1$ is completely divisible by a whole number. Which of the following is completely divisible by the same number?



- (a) $2^{15} + 1$ (b) 5×2^{30}
 (c) $2^{90} + 1$ (d) $2^{60} + 1$

Q.47. Which of the following numbers will completely divide $7^{81} + 7^{82} + 7^{83}$?

- (a) 399 (b) 389 (c) 387 (d) 397

Q.48. When a positive integer is divided by d, the remainder is 15. When ten times of the same number is divided by d, the remainder is 6. The least possible value of d is:

- (a) 9 (b) 12 (c) 16 (d) 18

Q.49. When 200 is divided by a positive integer x, the remainder is 8. How many values of x are there?

- (a) 7 (b) 5 (c) 8 (d) 6

Q.50. How many natural numbers less than 1000 are divisible by 5 or 7 but NOT by 35?

- (a) 285 (b) 313 (c) 341 (d) 243

Q.51. If the six digit number 479xyz is exactly divisible by 7, 11 and 13, then $\{(y + z) \div x\}$ is equal to:

- (a) $\frac{11}{9}$ (b) 4 (c) $\frac{13}{7}$ (d) $\frac{7}{13}$

Q.52. Which among the following is the smallest?

- (a) $\sqrt{401} - \sqrt{399}$ (b) $\sqrt{101} - \sqrt{99}$
 (c) $\sqrt{301} - \sqrt{299}$ (d) $\sqrt{201} - \sqrt{199}$

Q.53. Two positive numbers differ by 2001. When the larger number is divided by the smaller number, the quotient is 9 and the remainder is 41. The sum of the digits of the larger number is:

- (a) 15 (b) 11 (c) 10 (d) 14

Q.54. When a two - digit number is multiplied by the sum of its digits, the product is 424. When the number obtained by interchanging its digits is multiplied by the sum of the digits, the result is 280. The sum of the digits of the given number is:

- (a) 6 (b) 9 (c) 8 (d) 7

Q.55. Let $x = (633)^{24} - (277)^{38} + (266)^{54}$. What is the unit digit of x?

- (a) 7 (b) 6 (c) 4 (d) 8

Q.56. The sum of the digits of a two-digit number is $\frac{1}{7}$ of the number. The unit digit is 4 less than the tens digit. If the number obtained on reversing its digit is divided by 7, the remainder will be:

- (a) 4 (b) 5 (c) 1 (d) 6

Q.57. A gardener planted 1936 saplings in a garden such that there were as many rows of saplings as the columns. The number of rows planted is:

- (a) 46 (b) 44 (c) 48 (d) 42

Q.58. What is the difference between the largest and smallest numbers of the four digits created using numbers 2, 9, 6, 5? (Each number can be used only once)

- (a) 6993 (b) 7056 (c) 6606 (d) 7083

Q.59. The square root of which of the following is a rational number?

- (a) 1250.49 (b) 6250.49
 (c) 1354.24 (d) 5768.28

Q.60. The value of $22.\overline{4} + 11.\overline{567} - 33.\overline{59}$ is

- (a) $0.\overline{32}$ (b) $0.\overline{412}$ (c) $0.\overline{31}$ (d) $0.\overline{412}$

Examination wise Questions

SSC CGL 2025 Tier - 1

Q.61. Arrange the fractions $\frac{6}{11}, \frac{5}{8}, \frac{4}{7}$ and $\frac{3}{5}$ in ascending order.

SSC CGL 12/09/2025 (Shift 1)

- (a) $\frac{3}{5}, \frac{4}{7}, \frac{5}{8}, \frac{6}{11}$ (b) $\frac{6}{11}, \frac{4}{7}, \frac{3}{5}, \frac{5}{8}$
 (c) $\frac{6}{11}, \frac{3}{5}, \frac{4}{7}, \frac{5}{8}$ (d) $\frac{6}{11}, \frac{4}{7}, \frac{5}{8}, \frac{3}{5}$

Q.62. Let $x = \sqrt{3} + \sqrt{5}$ and $y = \sqrt{8} + \sqrt{3}$. Which is greater?

SSC CGL 12/09/2025 (Shift 2)

- (a) $x = y$ (b) $x > y$
 (c) $x < y$ (d) Cannot be determined

Q.63. Suppose $x = \sqrt{2} + \sqrt{3}$, $y = \sqrt{6} + \sqrt{1}$, and $z = 3\sqrt{2} + 1$; then which of the following is true?

SSC CGL 13/09/2025 (Shift 1)

- (a) $x < y < z$ (b) $z < y < x$
 (c) $y < x < z$ (d) $x < z < y$

Q.64. If $\left(\frac{1}{4 - \sqrt{15}}\right) - \left(\frac{1}{\sqrt{15} - \sqrt{14}}\right) + \left(\frac{1}{\sqrt{14} - \sqrt{13}}\right) - \left(\frac{1}{\sqrt{13} - \sqrt{12}}\right) + \left(\frac{1}{\sqrt{12} - \sqrt{11}}\right) - \left(\frac{1}{\sqrt{11} - \sqrt{10}}\right) + \left(\frac{1}{\sqrt{10} - 3}\right) = x$

Then find the value of $(x + 1)$?

SSC CGL 13/09/2025 (Shift 2)

- (a) 8 (b) 7 (c) 2 (d) 0

Q.65. If $a = \sqrt{5} - \sqrt{3}$, and $b = \sqrt{3} - \sqrt{2}$, then which of the following is true?

SSC CGL 14/09/2025 (Shift 3)

- (a) $a = b$ (b) $a < b$
 (c) $a > b$ (d) Cannot be determined

Q.66. Which of the following is equal to 14?

SSC CGL 15/09/2025 (Shift 2)

- (a) $(\sqrt{11} + \sqrt{5})^2 - 2\sqrt{55}$

- (b) $(\sqrt{5} + \sqrt{5})^2$
 (c) $(\sqrt{2} + \sqrt{3})^2 + 2\sqrt{6}$
 (d) $(\sqrt{8} + \sqrt{6})^2 - 2\sqrt{48}$

Q.67. Solve system:

$$y = 4x + 2 \text{ and } y = -2x + 8$$

SSC CGL 15/09/2025 (Shift 2)

- (a) (1, 6) (b) (2, 5) (c) (3, 7) (d) (0, 1)

Q.68. Express 0.232323... as a fraction.

SSC CGL 15/09/2025 (Shift 2)

- (a) $\frac{23}{99}$ (b) $\frac{23}{90}$ (c) $\frac{23}{999}$ (d) $\frac{23}{900}$

Q.69. Multiply 0.48 by 0.7 and express the result as a decimal.

SSC CGL 17/09/2025 (Shift 3)

- (a) 0.386 (b) 0.336 (c) 0.236 (d) 0.254

Q.70. Identify the incorrect relationship(s) from the list below:

(i) $\sqrt{5} + \sqrt{6} = \sqrt{7} + \sqrt{4}$

(ii) $\sqrt{5} + \sqrt{6} < \sqrt{7} + \sqrt{4}$

(iii) $\sqrt{5} + \sqrt{6} > \sqrt{7} + \sqrt{4}$

SSC CGL 18/09/2025 (Shift 2)

- (a) (ii) (b) (i)
 (c) (i) and (ii) (d) (ii) and (iii)

Q.71. Which of the following is correct?

(i) $\sqrt{11} + \sqrt{3} > \sqrt{12} + \sqrt{2}$

(ii) $\sqrt{11} + \sqrt{3} < \sqrt{12} + \sqrt{2}$

(iii) $\sqrt{11} + \sqrt{3} = \sqrt{12} + \sqrt{2}$

SSC CGL 19/09/2025 (Shift 1)

- (a) (i) (b) (ii) (c) (iii) (d) (i), (ii) and (iii)

Q.72. Which of the following is the least 6-digit number, which is not a perfect square?

SSC CGL 19/09/2025 (Shift 2)

- (a) 100489 (b) 100000
 (c) 101124 (d) 100788

Q.73. If you subtract the square of a number from 2796 and then multiply the result by 13, you get 35711. What is that number?

SSC CGL 19/09/2025 (Shift 3)

- (a) ± 8 (b) ± 6 (c) ± 9 (d) ± 7

Q.74. Choose the correct relation:

(i) $\sqrt{9} < 3.1$

(ii) $\sqrt{15} < \sqrt{17}$

(iii) $\sqrt{8} < \sqrt{11}$

SSC CGL 19/09/2025 (Shift 3)

- (a) (i) only (b) (i), (ii) and (iii)
 (c) (ii) and (iii) (d) (i) and (ii)

Q.75. What type of number is obtained when a rational number is added to an irrational number?

SSC CGL 20/09/2025 (Shift 1)

- (a) Always Rational
 (b) Always Irrational



- (c) Always Integer
(d) Sometimes Rational, Sometimes Irrational

Q.76. Find the solution to the system:

$$y = 7x - 4 \text{ and } y = -2x + 14.$$

SSC CGL 20/09/2025 (Shift 1)

- (a) (2, 10) (b) (17, 2)
(c) (1.5, 3.5) (d) (1.5, -4.5)

Q.77. Which of these is both an Integer and a Whole number?

SSC CGL 20/09/2025 (Shift 2)

- (a) -6 (b) 1 (c) 4.5 (d) $\frac{3}{2}$

Q.78. Find the value of x in the given

$$\text{equation: } \frac{\sqrt{8+x} + \sqrt{8-x}}{\sqrt{8+x} - \sqrt{8-x}} = 5$$

SSC CGL 20/09/2025 (Shift 3)

- (a) $\frac{46}{13}$ (b) $\frac{36}{13}$ (c) $\frac{40}{13}$ (d) $\frac{66}{17}$

Q.79. Which of the following numbers is an integer but not a natural number ?

SSC CGL 20/09/2025 (Shift 3)

- (a) 2 (b) 0 (c) 7 (d) 9

Q.80. Which of the following is a perfect square and an Integer ?

SSC CGL 21/09/2025 (Shift 1)

- (a) Square root of 7 (b) Square root of 16
(c) 4.3 (d) 5.4

Q.81. What is the smallest number that must be added to 5600 to make it a perfect square ?

SSC CGL 21/09/2025 (Shift 1)

- (a) 23 (b) 24 (c) 25 (d) 26

Q.82. Which of the following numbers belongs to the set of Integers but NOT to the set of Whole Numbers?

SSC CGL 21/09/2025 (Shift 2)

- (a) 0 (b) $\frac{1}{2}$ (c) -5 (d) +5

Q.83. Which of these is a number that is real but NOT rational?

SSC CGL 21/09/2025 (Shift 3)

- (a) $\frac{3}{5}$ (b) 7 (c) Square root of 5 (d) -3

Q.84. A water tub can hold 12.5 liters of water. How many such water tubs are needed to fill a 1 cubic meter tank ?

SSC CGL 22/09/2025 (Shift 1)

- (a) 80 (b) 70 (c) 65 (d) 90

Q.85. The decimal 0.785 is best classified as:

SSC CGL 22/09/2025 (Shift 2)

- (a) Whole Number (b) Irrational Number
(c) Integer (d) Rational Number

Q.86. Which of the following numbers is both a Rational Number and a

Terminating Decimal ?

SSC CGL 22/09/2025 (Shift 3)

- (a) $\frac{2}{3}$ (b) $\frac{7}{5}$ (c) $\sqrt{5}$ (d) π

Q.87. A recipe requires 0.85 liters of milk for 15 servings. How many servings can be made with 4.25 liters of milk ?

SSC CGL 22/09/2025 (Shift 3)

- (a) 78 (b) 70 (c) 72 (d) 75

Q.88. A number is such that if you square it and then subtract Fifth time the number, the result is 150. What is the number ?

SSC CGL 22/09/2025 (Shift 3)

- (a) 10 (b) 15 (c) 20 (d) 25

Q.89. The set of **integers** includes which of the following?

SSC CGL 23/09/2025 (Shift 1)

- (a) Only Positive Numbers
(b) Both Positive and Negative Numbers
(c) Only Whole Numbers
(d) Only Natural Numbers

Q.90. Which of these numbers is neither Rational nor Integer ?

SSC CGL 23/09/2025 (Shift 3)

- (a) $\sqrt{3}$ (b) -5 (c) 0 (d) 4.5

Q.91. Which of the following statements is CORRECT about Whole Numbers?

SSC CGL 24/09/2025 (Shift 2)

- (a) Whole numbers are the same as natural numbers.
(b) Whole numbers include all positive integers and zero.
(c) Whole numbers include both positive and negative integers.
(d) Whole numbers include all fractions and decimals.

Q.92. Which set includes numbers like -7, 4, and 0 ?

SSC CGL 25/09/2025 (Shift 1)

- (a) Natural Numbers (b) Whole Numbers
(c) Irrational Numbers (d) Integers

Q.93. Which set does contain non zero positive numbers ?

SSC CGL 25/09/2025 (Shift 3)

- (a) Integers
(b) Real Numbers
(c) Rational Numbers
(d) Natural Numbers

Q.94. A number is increased by 100, and the result becomes a perfect square. The original number lies between two consecutive square numbers, 729 and 784. Find the original number.

SSC CGL 14/10/2025 (Shift 2)

- (a) 700 (b) 732 (c) 741 (d) 745

Q.95. Which of the following numbers is an integer but not a natural number ?

SSC CGL 14/10/2025 (Shift 3)

- (a) 3 (b) 0 (c) 6 (d) 9

SSC Selection Post (Phase - XIII)

Q.96. Which number set includes all rational and irrational numbers?

Matriculation Level 24/07/2025 (Shift - 1)

- (a) Natural numbers (b) Whole numbers
(c) Real numbers (d) Integer numbers

Q.97. What is $\frac{147}{22}$ as a mixed number ?

Matriculation Level 24/07/2025 (Shift - 1)

- (a) $6\frac{15}{22}$ (b) $5\frac{17}{22}$
(c) $6\frac{17}{22}$ (d) $7\frac{15}{22}$

Q.98. How many natural numbers are there between 13^2 and 15^2 ?

Matriculation Level 24/07/2025 (Shift - 1)

- (a) 30 (b) 45 (c) 55 (d) 50

Q.99. Simplify the following expression and identify its nature:

$$\frac{(7\sqrt{5} \times \sqrt{20})}{\sqrt{4}}$$

Matriculation Level 25/07/2025 (Shift - 1)

- (a) $14\sqrt{5}$, irrational number
(b) 35, rational number
(c) $10\sqrt{2}$, irrational number
(d) $28\sqrt{5}$, irrational number

Q.100. Irrational numbers are a subset of:

Matriculation Level 25/07/2025 (Shift - 1)

- (a) Integers (b) Whole numbers
(c) Real numbers (d) Natural numbers

Q.101. The square root of 196 is:

Matriculation Level 25/07/2025 (Shift - 1)

- (a) 12 (b) 14 (c) 13 (d) 15

Q.102. Which of the following is true?

Matriculation Level 26/07/2025 (Shift - 1)

- (a) All natural numbers are integers.
(b) All irrational numbers are rational.
(c) All real numbers are integers.
(d) No integer is rational.

Q.103. What is the square of 73?

Matriculation Level 26/07/2025 (Shift - 1)

- (a) 5229 (b) 5529 (c) 5429 (d) 5329

Q.104. $\sqrt{6} + \sqrt{10}$ is classified as:

Matriculation Level 28/07/2025 (Shift - 1)

- (a) Whole Number (b) Integer
(c) Rational (d) Irrational



Q.105. Fill in the blank: A number that cannot be expressed as a fraction is called ____.

Matriculation Level 28/07/2025 (Shift - 1)

- (a) Whole (b) Integer
(c) Rational (d) Irrational

Q.106. 4.75 as a mixed number is:

Matriculation Level 28/07/2025 (Shift - 1)

- (a) $4\frac{1}{2}$ (b) $4\frac{3}{4}$ (c) $4\frac{2}{5}$ (d) $4\frac{1}{3}$

Q.107. What is the unit digit (ones place) of the square of 67?

Matriculation Level 28/07/2025 (Shift - 1)

- (a) 4 (b) 7 (c) 9 (d) 3

Q.108. Which of the following statements is true?

Matriculation Level 29/07/2025 (Shift - 1)

- (a) All integers are real.
(b) All real numbers are whole.
(c) All whole numbers are irrational.
(d) No integers are rational.

Q.109. The relationship between natural numbers and whole numbers can be expressed as:

Matriculation Level 29/07/2025 (Shift - 1)

- (a) All natural numbers are whole numbers
(b) All whole numbers are natural numbers
(c) Natural numbers and whole numbers are the same set
(d) Natural numbers and whole numbers have no elements in common

Q.110. Which of the following numbers is a perfect square?

Matriculation Level 29/07/2025 (Shift - 1)

- (a) 1849 (b) 2056 (c) 3991 (d) 2507

Q.111. Which of the following is not an integer?

Matriculation Level 30/07/2025 (Shift - 1)

- (a) $\frac{(2^3 - 2)}{2}$ (b) $\sqrt{121}$
(c) $\frac{3}{\sqrt{3}}$ (d) $\frac{(7^2 - 1)}{2}$

Q.112. Which of the following pairs of number sets represent exactly the same collection of numbers?

Matriculation Level 30/07/2025 (Shift - 1)

- (a) Even natural numbers and positive multiples of 2
(b) Whole numbers and negative integers
(c) Prime numbers and composite numbers
(d) Rational numbers and integers

Q.113. What is the sum of $2\frac{3}{7} + 5\frac{4}{9}$

written as a mixed number in simplest form?

Matriculation Level 30/07/2025 (Shift - 1)

- (a) $4\frac{5}{21}$ (b) $5\frac{2}{7}$ (c) $6\frac{1}{9}$ (d) $7\frac{55}{63}$

Q.114. A wire measures $18\frac{2}{5}$ meters in length. It is divided into smaller pieces, each 1.6 meters long. How many complete pieces can be obtained, and what length of wire will be left unused?

Matriculation Level 30/07/2025 (Shift - 1)

- (a) 11 pieces, 0.8 m left
(b) 15 pieces, 1.5 m left
(c) 11 pieces, 0.75 m left
(d) 13 pieces, 0.25 m left

Q.115. Find the square root of 73441

Matriculation Level 30/07/2025 (Shift - 1)

- (a) 271 (b) 272 (c) 265 (d) 268

Q.116. What type of number is the sum of $\sqrt{5}$ and $(3 - \sqrt{5})$?

Matriculation Level 31/07/2025 (Shift - 1)

- (a) an irrational number
(b) a rational number
(c) neither rational nor irrational
(d) cannot be determined

Q.117. Which number is added in addition to the natural numbers to make the set of whole numbers larger than the natural numbers?

Matriculation Level 01/08/2025 (Shift - 1)

- (a) 1 (b) -1 (c) 0 (d) $\frac{1}{2}$

Q.118. What types of numbers can each point to the right of 0 on the number line represent?

Matriculation Level 01/08/2025 (Shift - 2)

- (a) Only natural numbers
(b) Only positive integers
(c) Positive real numbers
(d) Only rational numbers

Q.119. If we look at the set of Integers and the set of Irrational Numbers, what elements are found in common in both the sets?

Matriculation Level 01/08/2025 (Shift - 3)

- (a) Only integers (b) Only irrational
(c) real numbers (d) No number

Q.120. How many two-digit numbers satisfy this property? The last digit (unit's digit) of the square of the two-digit number is 9.

Matriculation Level 01/08/2025 (Shift - 3)

- (a) 12 (b) 14 (c) 15 (d) 18

Q.121. If a number is irrational, it cannot be:

Matriculation Level 02/08/2025 (Shift - 1)

- (a) rational (b) real
(c) positive (d) negative

Q.122. Which is the number group that includes all rational and irrational numbers?

Matriculation Level 29/08/2025 (Shift - 3)

- (a) integer (b) real numbers
(c) Natural numbers (d) whole numbers

Q.123. Which of the following squares will end with the digit 4?

Matriculation Level 29/08/2025 (Shift - 3)

- (a) 18^2 (b) 27^2 (c) 23^2 (d) 31^2

Q.124. Which of the following is a perfect square of an even number?

Higher Secondary 24/07/2025 (Shift - 2)

- (a) 625 (b) 225 (c) 289 (d) 256

Q.125. Two numbers 89032 and 84179 are divided by a certain three-digit number and leave the same remainder. What is the sum of the digits of the divisor?

Higher Secondary 24/07/2025 (Shift - 2)

- (a) 5 (b) 4 (c) 6 (d) 7

Q.126. Which of the following sets is a proper subset of the set of whole numbers?

Higher Secondary 24/07/2025 (Shift - 2)

- (a) Natural numbers (b) Integers
(c) Rational numbers (d) Real numbers

Q.127. Match the sets in Column I with their correct descriptions in Column II:

Column I	Column II
A. Natural numbers (N)	1. Only positive counting numbers
B. Integers (Z)	2. All positive, negative numbers and zero
C. Even numbers (E)	3. The set of numbers divisible by 2
D. Real numbers (R)	4. The collection of all rational and irrational numbers

Higher Secondary 24/07/2025 (Shift - 2)

- (a) A-1, B-2, C-3, D-4 (b) A-3, B-1, C-4, D-2
(c) A-4, B-3, C-1, D-2 (d) A-2, B-4, C-3, D-1

Q.128. If $x = \sqrt{7}$ and $y = 5 - \sqrt{7}$, then which of the following is true?

Higher Secondary 25/07/2025 (Shift - 2)

- (a) $x + y$ is irrational (b) $x + y$ is rational
(c) $x - y$ is rational (d) $x - y$ is integer

Q.129. Consider the following two statements, labeled Assertion (A) and Reason (R). Evaluate them and choose the appropriate answer from the given options.

Assertion (A): Every positive integer can



be expressed uniquely as a product of powers of prime numbers.

Reason (R): In the prime factorization of any integer, the prime numbers used and their respective powers are unique.

Higher Secondary 25/07/2025 (Shift - 2)

- (a) Both A and R are true, and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true, but R is false.
 (d) A is false, but R is true.

Q.130. Which of the following numbers cannot be the square of a natural number?

Higher Secondary 25/07/2025 (Shift - 2)

- (a) 35344 (b) 4225 (c) 1156 (d) 2118

Q.131. What is the smallest positive integer such that its product with 180 is a perfect square ?

Higher Secondary 25/07/2025 (Shift - 2)

- (a) 7 (b) 5 (c) 4 (d) 10

Q.132. A student claims that the sum of two irrational numbers is always irrational. Which of the following pair of numbers disproves this claim ?

Higher Secondary 26/07/2025 (Shift - 2)

- (a) $\sqrt{3} + \sqrt{3}$ (b) $\sqrt{5} + \sqrt{3}$
 (c) $\pi + \sqrt{3}$ (d) $\sqrt{5} + (-\sqrt{5})$

Q.133. Which of the following numbers is an integer but not natural ?

Higher Secondary 26/07/2025 (Shift - 2)

- (a) 8 (b) -3 (c) 5 (d) 12

Q.134. A positive number exceeds its square root by 210 . Find the number given that it is also a multiple of 15.

Higher Secondary 28/07/2025 (Shift - 2)

- (a) 225 (b) 256 (c) 400 (d) 625

Q.135. Given below are two statements marked as Assertion (A) and Reason (R). Read and mark the correct options:

Assertion (A):

The sum of any two even natural numbers is always even.

Reason (R):

If a number can be written as $2k$, then it is even, and the sum of two even numbers can be written as $2(k_1 + k_2)$, which is even.

Higher Secondary 28/07/2025 (Shift - 2)

- (a) Both A and R are true, and R is the correct explanation of A
 (b) Both A and R are true, but R is not the correct explanation of A
 (c) A is true, but R is false
 (d) A is false, but R is true

Q.136. Which of the following relationship between the sets of

integers(z), rational numbers(x), and real numbers(y) holds good?

Higher Secondary 29/07/2025 (Shift - 2)

- (a) $x \subseteq z \subseteq y$ (b) $y \subseteq x \subseteq z$
 (c) $z \subseteq y \subseteq x$ (d) $z \subseteq x \subseteq y$

Q.137. If a pipe of length $\sqrt{392}$ m is cut into 14 equal parts, what is the length of each part, and is it rational or irrational?

Higher Secondary 30/07/2025 (Shift - 2)

- (a) Each part is $\sqrt{4}$ m irrational
 (b) Each part is $\sqrt{2}$ m irrational
 (c) Each part is 7 m rational
 (d) Each part is $\sqrt{11}$ m irrational

Q.138. If p is a natural number, then $p(p+1)(p+2)(p+3)+1$ is always:

Higher Secondary 30/07/2025 (Shift - 2)

- (a) A perfect cube
 (b) Not a perfect square
 (c) A perfect square
 (d) An odd number

Q.139. The binary number 100010 in the decimal number system is equal to:

Higher Secondary 30/07/2025 (Shift - 2)

- (a) 33 (b) 34 (c) 35 (d) 36

Q.140. Which of the following is not a like surd?

Higher Secondary 31/07/2025 (Shift - 2)

- (a) $-2\sqrt{15}$ (b) $-3\sqrt{7}$ (c) $5\sqrt{37}$ (d) $\sqrt{25}$

Q.141. The number of natural numbers between $\sqrt{80}$ and $\sqrt{800}$ is:

Higher Secondary 31/07/2025 (Shift - 2)

- (a) 10 (b) 35 (c) 25 (d) 20

Q.142. The number of digits in the square root of a perfect square containing $4n$ digits is:

Higher Secondary 31/07/2025 (Shift - 2)

- (a) $2n$ (b) $2n-1$ (c) $2n+1$ (d) $4n$

Q.143. When the binary value 10110101 is added to 11001010, the resulting sum in decimal form is:

Higher Secondary 31/07/2025 (Shift - 2)

- (a) 330 (b) 331 (c) 332 (d) 383

Q.144. Consider the number $\sqrt{(-49)}$.

Which of the following statements is true about the above number ?

Higher Secondary 31/07/2025 (Shift - 2)

- (a) It represents a real number and is also an integer.
 (b) It does not belong to the set of real numbers but belongs to the set of complex numbers.
 (c) It is a real number though it is not an integer.
 (d) It is neither a real number nor a complex number.

Q.145. Which of the following square roots is rational ?

Higher Secondary 02/08/2025 (Shift - 2)

- (a) $\sqrt{144}$ (b) $\sqrt{125}$ (c) $\sqrt{51}$ (d) $\sqrt{200}$

Q.146. Arrange the following numbers in ascending order: $\sqrt{2}, \sqrt[4]{3}, \sqrt[3]{7}$

Higher Secondary 29/08/2025 (Shift - 2)

- (a) $\sqrt[3]{7} < \sqrt{2} < \sqrt[4]{3}$ (b) $\sqrt[3]{7} < \sqrt[4]{3} < \sqrt{2}$
 (c) $\sqrt[4]{3} < \sqrt{2} < \sqrt[3]{7}$ (d) $\sqrt[4]{3} < \sqrt[3]{7} < \sqrt{2}$

Q.147. Which of the following numbers is a perfect square and a multiple of 5?

Higher Secondary 29/08/2025 (Shift - 2)

- (a) 225 (b) 250 (c) 215 (d) 125

Q.148. Which of the following numbers is an integer?

Higher Secondary 29/08/2025 (Shift - 2)

- (a) 7.28318 (b) -15 (c) $\frac{5}{3}$ (d) $\sqrt{15}$

Q.149. The binary number 1111111 divided by 11 gives_____ in the decimal system.

Higher Secondary 29/08/2025 (Shift - 2)

- (a) 42.33 (b) 42.8 (c) 44.33 (d) 42.5

Q.150. The unit digit of 11^{203} is:

Graduate Level 25/07/2025 (Shift - 3)

- (a) 5 (b) 2 (c) 0 (d) 1

Q.151. What type of number is obtained when two irrational numbers are added ?

Graduate Level 26/07/2025 (Shift - 3)

- (a) Always Rational
 (b) Always Irrational
 (c) Sometimes Rational, Sometimes Irrational
 (d) Always Integer

Q.152. What is the sum of all three-digit numbers divisible by 8 ?

Graduate Level 31/07/2025 (Shift - 3)

- (a) 61376 (b) 62707 (c) 62006 (d) 62407

Q.153. Which is the most significant 4-digit number exactly divisible by 77 ?

Graduate Level 02/08/2025 (Shift - 3)

- (a) 9999 (b) 9977 (c) 9960 (d) 9933

eduquity-based pattern (ebp) Questions

Q.154. A number 37 is divided into two parts such that 5 times one part and 11 times the other is together equal to 227. Find the value of 9 times the smaller number.

- (a) 60 (b) 61 (c) 62 (d) 63

Q.155. The number of heads on the farm, as determined by the farmer, was 60. When he counted, he discovered there were 150 legs. If all that was present



were pigeons or horses, and every horse on the property had four legs while every pigeon had two, how many horses were there on the property?

- (a) 14 (b) 12 (c) 15 (d) 10

Q.156. The length of a violin string varies inversely with respect to the frequency of its vibrations. The frequency of a 14-inch-long violin string is 450 cycles per second. Find the frequency of a 12-inch violin string.

- (a) 515 cycles per second
(b) 525 cycles per second
(c) 520 cycles per second
(d) 510 cycles per second

Q.157. b is the sum of two quantities, one of which varies inversely as the square of a^2 , and the other of which varies directly as a . Identify the relationship between a and b if $b = 49$ when $a = 3$ or 5 .

- (a) $a = 3b$ (b) $a = 3b^2 - 7/a$
(c) $b = 8a + 225/a^2$ (d) $b = 2a$

Q.158. If p is directly proportional to q and is $p = 7$ when $q = 6$, then what is the value of p when $q = 21$?

- (a) 24.0 (b) 24.5 (c) 23.5 (d) 25.0

Q.159. If P ($P > 0$) is the least possible number that must be subtracted from 7, 9, and 12 so that the resulting numbers are in continued proportion, then find the value of $P^2 + 1$.

- (a) 10 (b) 12 (c) 3 (d) 8

Q.160. Which of the following can be the value of ' k ' so that the number 217924 k is divisible by 6?

- (a) 4 (b) 6 (c) 2 (d) 0

Q.161. Without doing the actual division, find the remainder when 28735429 is divided by 9.

- (a) 4 (b) 2 (c) 8 (d) 9

Q.162. If 7845K854 is divisible by 11, then what is the value of K ?

- (a) 6 (b) 9 (c) 7 (d) 8

Q.163. Which of the following is the largest 5-digit number divisible by 47?

- (a) 99999 (b) 98888 (c) 99969 (d) 10000

Q.164. Akshay purchased pens and notebooks from a shop for a total sum of ₹180. From the following data, what is the value of ' X '?

Items	Price per unit (in ₹)	Quantity
Pens	10	X
Notebooks	20	6

- (a) 4 (b) 6 (c) 5 (d) 7

Q.165. If $(77^{77} + 77)$ is divided by 78, then the remainder will be:

- (a) 76 (b) 77 (c) 78 (d) 1

Q.166. Which digits should come in the place of $*$ and $\#$, respectively, if the 7-digit number 62684 $*$ $\#$ is divisible by both 8 and 5?

- (a) 2 and 0 (b) 4 and 0
(c) 5 and 0 (d) 0 and 5

Q.167. Find the smallest number to be added to 999, so that 99 divides the sum exactly.

- (a) 99 (b) 0 (c) 90 (d) 9

Q.168. In a division problem, the divisor is 4 times the quotient and 2 times the remainder. If the remainder is 32, then find the dividend.

- (a) 1056 (b) 1650 (c) 3240 (d) 1065

Q.169. Which of the following numbers is divisible by both 2 and 5?

- (a) 63840 (b) 20792 (c) 37915 (d) 37254

Q.170. Find the quotient, when the mean proportional of $7\frac{1}{5}$ and 245 is divided by an even prime number

- (a) 49 (b) 42 (c) 21 (d) 36

Q.171. Naman bought a few apples for ₹720 from a shop. He negotiated the price and the shopkeeper reduced it by ₹2 per apple. Due to this Naman could buy four more apples than what he had bought earlier. How many apples did he originally buy?

- (a) 48 (b) 44 (c) 36 (d) 40

Q.172. Replace $*$ by the smallest digit so that $723*56*$ is divisible by 6

- (a) 4 (b) 3 (c) 2 (d) 1

Q.173. In a garden, there are 6 daisy plants the first year. Each year, a gardener adds 3 new daisy plants. He has 26 jasmine plants the first year and loses 2 each year. When will the number of daisy plants equal to the number of jasmine plants after the first year?

- (a) 5 years (b) 4 years
(c) 2 years (d) 6 years

Q.174. Which of the following numbers is divisible by 2, 5, 10 and 11?

- (a) 203467 (b) 830942
(c) 589270 (d) 1234560

Q.175. Find the smallest value that must be assigned to number ' a ' in order for 91876a2 to be divisible by 8.

- (a) 3 (b) 0 (c) 1 (d) 2

Q.176. If $321y72$ is a multiple of 6, where y is a digit, what is the least value of y ?

- (a) 3 (b) 0 (c) 7 (d) 2

Q.177. At a cafe, Rama bought drinks for her 7 friends at ₹240. Some of them wanted coffee, while others wanted black tea. A regular cup of coffee cost ₹40 and a regular cup of black tea cost ₹30. How many cups of black tea did she purchase?

- (a) 5 (b) 4 (c) 3 (d) 6

Q.178. Find the remainder: $(17^{13} - 21) \div 18$

- (a) 15 (b) 17 (c) 14 (d) 21

Q.179. Find the least value of x for which $57x716$ is divisible by 9.

- (a) 1 (b) 9 (c) 0 (d) 8

Q.180. If P is a digit such that $6954P$ is divisible by 11, then P equals:

- (a) 2 (b) 8 (c) 6 (d) 7

Q.181. When $f(m) = m^5 + 5m^4 - 3m + 7$ is divided by $(m - 2)$, then the remainder is

- (a) 7 (b) 5 (c) 0 (d) 113

Q.182. A number r when divided by 8 leaves remainder 3. What will be the remainder when $(r^2 + 6r + 7)$ is divided by 8?

- (a) 2 (b) 3 (c) 1 (d) 4

Q.183. Find the remainder, if 19^{200} is divided by 20.

- (a) 2 (b) -1 (c) 1 (d) 3

Q.184. Which of the following fractions is the largest?

$$\frac{2}{3}, \frac{6}{53}, \frac{79}{90}, \frac{33}{44}$$

- (a) $\frac{2}{3}$ (b) $\frac{33}{44}$ (c) $\frac{6}{53}$ (d) $\frac{79}{90}$

Q.185. If the 9-digit number $5x79856y6$ is divisible by 36, then what is the negative value of $\sqrt{(2x + y)}$ for the largest possible value of y , given x and y are natural numbers?

- (a) -7 (b) -2 (c) -4 (d) -5

Q.186. Anna and Ben each have a collection of marbles. If Anna gives one marble to Ben, they will both have an equal number of marbles. Conversely, if Ben gives one marble to Anna, she will end up with three times the number of marbles that Ben has. What is the total number of marbles they both have?

- (a) 10 (b) 8 (c) 11 (d) 9

Q.187. What number should be subtracted from $x^3 - 4x^2 - 8x + 11$ to make the number divisible by $(x + 2)$?

- (a) 9 (b) 5 (c) 3 (d) 7

Q.188. Each of Ravi and Kavita had some marbles. Kavita had 12 more marbles than Ravi had. If each of them had one



more marble, then three times the number of marbles Kavita would then have had would have been equal to four times the number of marbles Ravi would then have had. How many marbles did Kavita actually have ?

- (a) 43 (b) 47 (c) 51 (d) 48

Q.189. From the given numbers A, B, C and D, which number is NOT divisible by 11?

A = 712712 B = 177210

C = 64614 D = 756148

- (a) C (b) D (c) B (d) A

Q.190. Which of the following numbers is NOT divisible by 8 ?

- (a) 7344 (b) 5544 (c) 7497 (d) 4608

Q.191. For what least value of n , $2^{2n} + 2n$ is completely divided by 6, where n is an integer?

- (a) 3 (b) 1 (c) 0 (d) 2

Q.192. The least number which must be subtracted from 7278745 so as to obtain a sum divisible by 11 is:

- (a) 3 (b) 1 (c) 5 (d) 2

Q.193. Ten chairs and six tables together cost ₹5,140; three chairs and two tables together cost ₹1,635. The cost of 1 chair and 1 table is:

- (a) ₹900 (b) ₹600 (c) ₹800 (d) ₹700

Q.194. The number 4,29,714 is divisible by:

- (a) 3 and 5 (b) 6 and 5
(c) 4 and 6 (d) 3 and 6

Q.195. Find the greatest value of 'k' in a 6-digit number 6745k2 such that the number is divisible by 3.

- (a) 8 (b) 6 (c) 9 (d) 7

Q.196. On dividing a number by 55, we get 28 as the remainder. On dividing the same number by 11, what is the remainder?

- (a) 5 (b) 8 (c) 7 (d) 6

Q.197. The ratio of two numbers is 2 : 5 and their difference is 210. The smaller number is _____.

- (a) 60 (b) 150 (c) 140 (d) 350

Q.198. If 436P5 is divisible by 3, then the largest value of P is:

- (a) 12 (b) 6 (c) 8 (d) 9

Q.199. Which composite number can divide the sum of the first 12 natural numbers?

- (a) 6 (b) 12 (c) 4 (d) 8

Q.200. If m is even, then $(8^m - 1)$ is divisible by :

- (a) 63 (b) 42 (c) 65 (d) 8

Q.201. Which of the following numbers is divisible by 6?

- (a) 128530 (b) 12581
(c) 438750 (d) 62233

Q.202. A number whose fifth part increased by 5 is equal to its third part decreased by 7. Find half of the number.

- (a) 45 (b) 150 (c) 80 (d) 60

Q.203. The sum of two consecutive even numbers is 174. Find the smaller number.

- (a) 84 (b) 90 (c) 86 (d) 88

Q.204. $7^{15} + 7^{16} + 7^{17}$ is divisible by which of the following given numbers?

- (a) 3 (b) 5 (c) 2 (d)

Q.205. Find the number when successively divided by 3, 5 and 7 leaves remainder 2, 1 and 3, respectively, and the last quotient is 3.

- (a) 367 (b) 365 (c) 362 (d) 360

Q.206. How many numbers lie between 2000 and 2020 that are divisible by 8 ?

- (a) 4 (b) 3 (c) 2 (d) 5

Q.207. On dividing a certain number by 304, we get 43 as the remainder. If the same number is divided by 16, what will be the remainder?

- (a) 8 (b) 11 (c) 15 (d) 12

Q.208. The sum of the five consecutive even numbers is 2720. The sum of the third and fifth numbers is :

- (a) 1392 (b) 1192 (c) 1092 (d) 1292

Q.209. What is the number of digits required for numbering a book with 428 pages ?

- (a) 1500 (b) 2000 (c) 988 (d) 1176

Q.210. An 8 digit number is divisible by 99. If the digits are shuffled, then the number is always divisible by _____.

- (a) 9 (b) 11 (c) 6 (d) 5

Q.211. When a number is divided by 512 it leaves a remainder 67. If the same number is divided by 32, then what will be the remainder?

- (a) 5 (b) 0 (c) 4 (d) 3

Q.212. Find the remainder when $179 \times 172 \times 173$ is divided by 17.

- (a) 2 (b) 9 (c) 3 (d) 6

Q.213. A seven-digit number 489y5z6 is divisible by 72. Which of the options gives the highest possible product of y and z ?

- (a) 3 (b) 42 (c) 30 (d) 21

Q.214. When the integer n is divided by 18, the quotient is x and the remainder is 6, When the integer n is divided by 25, the quotient is y and the remainder is 15. Which of the following is true?

- (a) $18x - 25y = 9$ (b) $18x + 25y = 9$
(c) $25y - 18x = 9$ (d) $25y + 18x = 9$

Q.215. Which of the following numbers is divisible by 22 ?

- (a) 654320 (b) 893002
(c) 602351 (d) 645372

Q.216. The number nearest to 7658 and exactly divisible by 45 is :

- (a) 7645 (b) 7660 (c) 7640 (d) 7650

Q.217. The sum of two-digit number and the number obtained by interchanging the digit is 77. If the difference of digits is 1, then the number is:

- (a) 67 (b) 34 (c) 12 (d) 45

Q.218. Write the smallest digit in the blank space of a number 7__7624, so that the number formed is divisible by 3.

- (a) 2 (b) 1 (c) 3 (d) 0

Q.219. k being any odd number greater than 1, $k^{33} - k$ is always divisible by:

- (a) 5 (b) 13 (c) 24 (d) 15

Q.220. How many numbers between 10 and 65 are divisible by 2, 3 and 4 ?

- (a) 5 (b) 7 (c) 9 (d) 3

Q.221. A's marks in Mathematics are directly proportional to practice time. In 6 hours of practice, A gets 70 marks. What should be the practice time (approximately) to get 90 Marks ?

- (a) 7.7 hours (b) 7 hours
(c) 8.3 hours (d) 8 hours

Q.222. A two - digit number is 12 more than five times the sum of its digits. The number formed by reversing the digits is 9 less than the original number. The number is:

- (a) 96 (b) 78 (c) 87 (d) 69

Q.223. When the integer n is divided by 5, the remainder is 3. What is the remainder if $6n$ is divided by 5 ?

- (a) 0 (b) 2 (c) 3 (d) 1

Q.224. Identify a single-digit number other than 1, which may exactly divide the number $17^3 + 18^3 - 16^3 - 15^3$

- (a) 3 (b) 5 (c) 7 (d) 2

Q.225. A natural number n divides 732 and leaves 12 as a remainder. How many values of n are possible?

- (a) 20 (b) 16 (c) 18 (d) 22

Q.226. The age of a tree depends on the number of rings in the trunk. The value of



a such tree is proportional to its age. If a tree with 40 rings and 50 years of age has a value of ₹50,000, then a tree with 30 rings is _____ years of age and its value is ₹_____.

- (a) 37.5, 57500 (b) 37.5, 37500
(c) 57.5, 37500 (d) 57.5, 57500

Q.227. ($8^6 + 1$) when divided by 7, would leave a remainder of:

- (a) 1 (b) 4 (c) 2 (d) 6

Q.228. The sum of the digits of a two-digit number is 9. If the digits are reversed, then the number is increased by 27. Find the number.

- (a) 63 (b) 36 (c) 54 (d) 45

Q.229. A number is divisible by 11 if and only if the difference of the sum of the digits in the even and odd positions in the number is:

- (a) 0 or multiple of 11 (b) 11 (c) 0 (d) 1

Q.230. When 2^{256} is divided by 17, the remainder would be:

- (a) 2 (b) 0 (c) 1 (d) 5

Q.231. The sum of two numbers is 76. Three times the larger number is 46 more than four times the smaller one. Find the two numbers.

- (a) 44 and 32 (b) 50 and 26
(c) 40 and 36 (d) 35 and 41

Q.232. If the number $6p5157q$ is divisible by 88, then $p \times q = \underline{\hspace{1cm}}$ where p and q are single digit numbers.

- (a) 12 (b) 18 (c) 20 (d) 15

Q.233. On dividing a certain number by 459, we get 19 as remainder. What will be the remainder, when the same number is divided by 17?

- (a) 11 (b) 1 (c) 13 (d) 2

Q.234. The sum of the digits of a two-digit number is 9. The number obtained by interchanging its digits exceeds the given number by 45, then the original number is:

- (a) 18 (b) 36 (c) 27 (d) 54

Q.235. Divide 25 into two parts such that 6 times of the larger part added to 4 times of the smaller part is equal to 130. Find the larger part.

- (a) 13 (b) 20 (c) 15 (d) 14

Q.236. The number 611611611611 is:

- (A) divisible by both 6 and 11
(B) neither divisible by 6 nor by 11
(C) divisible by 6 only
(D) divisible by 11 only
(a) (B) (b) (D) (c) (A) (d) (C)

Q.237. Arrange the following ratios in ascending order.

- 3 : 4, 3 : 5, 2 : 11, 7 : 2
(a) $3 : 5 < 2 : 11 < 3 : 4 < 7 : 2$
(b) $7 : 2 < 3 : 4 < 3 : 5 < 2 : 11$
(c) $2 : 11 < 3 : 5 < 3 : 4 < 7 : 2$
(d) $3 : 4 < 3 : 5 < 2 : 11 < 7 : 2$

Q.238. 18 students of a class took part in a quiz. If the number of girls is 8 more than the number of boys, find the product of the number of boys and girls.

- (a) 56 (b) 36 (c) 63 (d) 65

Q.239. Which of the following numbers is divisible by 11?

- (a) 908781 (b) 969331
(c) 701611 (d) 611571

Q.240. The number 10000 is exactly divisible by which of the following numbers?

- (a) 18 (b) 17 (c) 25 (d) 14

Q.241. If the number $3727x4$ is completely divisible by 8, then the smallest integer in place of x will be

- (a) 6 (b) 8 (c) 2 (d) 0

Q.242. Which of the following is a condition of divisibility of a number by six?

- (a) Sum of digits is divisible by 6
(b) Last digit is 3 or 6
(c) Sum of digits is divisible by 3 and last digit is even
(d) Sum of digits is divisible by 3 and first digit is even

Q.243. The largest 4 - digit number that is exactly divisible by 88 is:

- (a) 9848 (b) 9988 (c) 8888 (d) 9768

Q.244. Find the least value digit which is assigned to x so that the number $1972x471$ is divisible by 9.

- (a) 3 (b) 5 (c) 4 (d) 2

Q.245. If a number is divisible by both 11 and 13, then it must be:

- (a) divisible by (11×13)
(b) divisible by 42
(c) divisible by $(11 + 13)$
(d) divisible by $(13 - 11)$

Q.246. If a 9-digit number $937X728Y6$ is divisible by 72, then one of the possible values of X + Y is:

- (a) 12 (b) 8 (c) 9 (d) 5

Q.247. A number when divided by a divisor leaves a remainder of 24. When twice the original number is divided by the same divisor, the remainder is 13. What is the value of the divisor?

- (a) 35 (b) 37 (c) 25 (d) 30

Q.248. If $(5)^{55}$ is divisible by 4, then the remainder is:

- (a) 5 (b) 1 (c) 3 (d) 2

Q.249. How many numbers less than 1,000 are multiples of both 7 and 11?

- (a) 13 (b) 11 (c) 12 (d) 10

Q.250. Which of the following numbers is divisible by 4.

- (i) 67920598 (ii) 618703592
(iii) 618703594 (iv) 67920590
(a) (i) (b) (iii) (c) (iv) (d) (ii)

Q.251. In a division sum, the divisor is 13 times the quotient and 6 times the remainder. If the remainder is 39, then the dividend is:

- (a) 4,800 (b) 4,576 (c) 4,251 (d) 4,240

Q.252. Which of the following numbers is divisible by 36?

- (a) 1542 (b) 96272 (c) 55512 (d) 8840

Q.253. The number 1254216 is divisible by which of the following numbers?

- (a) 8 (b) 11 (c) 5 (d) 16

Q.254. Find the sum of $3 + 3^2 + 3^3 + \dots + 3^8$.

- (a) 9840 (b) 6561 (c) 6560 (d) 3280

Q.255. The number 2918245 is divisible by which of the following numbers?

- (a) 3 (b) 9 (c) 11 (d) 12

Q.256. $6^{25} + 6^{26} + 6^{27} + 6^{28}$ is divisible by:

- (a) 254 (b) 259 (c) 256 (d) 255

Q.257. If a car covers 90 km using 4 litres of petrol, how much distance (in km) will it cover using 32 litres of petrol?

- (a) 720 (b) 710 (c) 680 (d) 650

Q.258. In a division sum, the divisor is 10 times the quotient and 5 times the remainder. If the remainder is 12, then what is the dividend?

- (a) 368 (b) 372 (c) 386 (d) 352

Q.259. Which of the following numbers will completely divide $4^{12} + 4^{13} + 4^{14} + 4^{15}$

- (a) 17 (b) 11 (c) 3 (d) 7

Q.260. Without actual division, find the remainder when 379843 is divided by 3.

- (a) 4 (b) 1 (c) 3 (d) 2

Q.261. If the six-digit number $15x1y2$ is divisible by 44, then the minimum value of (x + y) is equal to:

- (a) 5 (b) 6 (c) 8 (d) 7

Q.262. If a 9-digit number $389x6378y$ is divisible by 72, then the value of $6x + 7y$ is:

- (a) 64 (b) 32 (c) 16 (d) 28

Q.263. Which of the following numbers is



divisible by 9?

- (a) 553986 (b) 350846
(c) 941201 (d) 132490

Q.264. The five-digit number 725yz is divisible by 15. What is the maximum possible value of the product of y and z
(a) 45 (b) 30 (c) 35 (d) 40

Q.265. If the 6 digit number 7002 * 4 is completely divisible by 8, then the smallest integer in place of * will be:
(a) 0 (b) 2 (c) 4 (d) 6

Q.266. If $72 * 72$ is divisible by 9, the missing * digit will be:
(a) 5 or 8 (b) 0 or 9 (c) 2 or 4 (d) 3 or 6

Q.267. The number 7918378 is divisible by:
(a) 13 (b) 9 (c) 11 (d) 4

Q.268. What least number must be subtracted from 2001 to get a number exactly divisible by 17?
(a) 12 (b) 13 (c) 11 (d) 14

Q.269. If * is a digit such that 7235 * is divisible by 11, then the value of * is:
(a) 9 (b) 6 (c) 5 (d) 8

Q.270. The average of 25 integers is zero. What is the maximum possible number of positive integers?
(a) 12 (b) 20 (c) 13 (d) 24

Q.271. Find the largest number which should replace P in the 7- digit number 87893P4 to make the number divisible by 4.
(a) 8 (b) 9 (c) 2 (d) 0

Q.272. Find the smallest non-zero value of k so that 7-digit number 48397k5 is divisible by 9
(a) 2 (b) 5 (c) 4 (d) 9

Q.273. What is the least number that must be added to the greatest 6-digit number so that the sum will be exactly divisible by 294?
(a) 234 (b) 194 (c) 269 (d) 189

Q.274. Which of the following sets is such that all its elements are divisors of the number 2520?
(a) 12, 49, 18 (b) 8, 9, 7
(c) 16, 15, 14 (d) 21, 10, 25

Q.275. What will be the greatest number 32a78b, which is divisible by 3 but NOT divisible by 9? (Where a and b are single digit numbers).
(a) 324781 (b) 329787
(c) 326787 (d) 329784

Q.276. The sum of the cubes of two given numbers is 10234, while the sum of the two given numbers is 34. What is

the positive difference between the cubes of the two given numbers?
(a) 3484 (b) 3488 (c) 3356 (d) 8602

Q.277. The six-digit number $N = 4a6b9c$ is divisible by 99, then the maximum sum of the digits of N is:
(a) 18 (b) 36 (c) 45 (d) 27

Q.278. What will be the remainder when $(265)^{4081} + 9$ is divided by 266?
(a) 8 (b) 6 (c) 1 (d) 9

Q.279. A and B have some toffees. If A gives one toffee to B, then they have an equal number of toffees. If B gives one toffee to A, then the toffees with A are double with B. The total number of toffees with A and B are _____.
(a) 12 (b) 10 (c) 14 (d) 15

Q.280. Find the smallest number that can be subtracted from 148109326 so that it becomes divisible by 8.
(a) 4 (b) 8 (c) 6 (d) 10

Q.281. The largest 5-digit number exactly divisible by 88 is:
(a) 99990 (b) 99984 (c) 99978 (d) 99968

Q.282. The average of squares of five consecutive odd natural numbers is 233. What is the average of the largest number and the smallest number?
(a) 11 (b) 17 (c) 13 (d) 15

Q.283. When $(2^{24} - 1)$ is divided by 7, the remainder is:
(a) 2 (b) 0 (c) 4 (d) 1

Q.284. When a positive integer n is divided by 12, the remainder is 5. What will be the remainder if $8n^2 + 7$ is divided by 12?
(a) 2 (b) 5 (c) 3 (d) 4

Q.285. The remainder when $75 \times 73 \times 78 \times 76$ is divided by 34 is:
(a) 18 (b) 12 (c) 22 (d) 15

Q.286. What is the least number of soldiers that can be drawn up in troops of 10, 12, 15, 18 and 20 soldiers, and also in form of a perfect square?
(a) 180 (b) 625 (c) 900 (d) 400

Answer Key :-

1.(b)	2.(c)	3.(b)	4.(a)
5.(c)	6.(b)	7.(b)	8.(b)
9.(c)	10.(c)	11.(d)	12.(a)
13.(d)	14.(b)	15.(c)	16.(a)
17.(d)	18.(a)	19.(d)	20.(c)

21.(d)	22.(b)	23.(a)	24.(d)
25.(a)	26.(d)	27.(d)	28.(d)
29.(d)	30.(a)	31.(d)	32.(a)
33.(b)	34.(d)	35.(b)	36.(c)
37.(d)	38.(b)	39.(c)	40.(b)
41.(a)	42.(a)	43.(c)	44.(b)
45.(b)	46.(d)	47.(a)	48.(c)
49.(c)	50.(a)	51.(b)	52.(a)
53.(d)	54.(c)	55.(d)	56.(d)
57.(b)	58.(d)	59.(c)	60.(d)
61.(b)	62.(c)	63.(a)	64.(a)
65.(c)	66.(d)	67.(a)	68.(a)
69.(b)	70.(c)	71.(a)	72.(d)
73.(d)	74.(b)	75.(b)	76.(a)
77.(b)	78.(c)	79.(b)	80.(b)
81.(c)	82.(c)	83.(c)	84.(a)
85.(d)	86.(b)	87.(d)	88.(b)
89.(b)	90.(a)	91.(b)	92.(d)
93.(d)	94.(c)	95.(b)	96.(c)
97.(a)	98.(c)	99.(b)	100.(c)
101.(b)	102.(a)	103.(d)	104.(d)
105.(d)	106.(b)	107.(c)	108.(a)
109.(a)	110.(a)	111.(c)	112.(a)
113.(d)	114.(a)	115.(a)	116.(b)
117.(c)	118.(c)	119.(d)	120.(d)
121.(a)	122.(b)	123.(a)	124.(d)
125.(b)	126.(a)	127.(a)	128.(b)
129.(a)	130.(d)	131.(b)	132.(d)
133.(b)	134.(a)	135.(a)	136.(d)
137.(b)	138.(c)	139.(b)	140.(d)
141.(d)	142.(a)	143.(d)	144.(b)
145.(a)	146.(c)	147.(a)	148.(b)
149.(a)	150.(d)	151.(c)	152.(a)
153.(d)	154.(d)	155.(c)	156.(b)
157.(c)	158.(b)	159.(a)	160.(c)
161.(a)	162.(b)	163.(c)	164.(b)
165.(a)	166.(b)	167.(c)	168.(a)
169.(a)	170.(c)	171.(c)	172.(c)
173.(b)	174.(c)	175.(a)	176.(b)
177.(b)	178.(c)	179.(a)	180.(a)
181.(d)	182.(a)	183.(c)	184.(d)
185.(d)	186.(b)	187.(c)	188.(b)
189.(b)	190.(c)	191.(b)	192.(b)
193.(d)	194.(d)	195.(c)	196.(d)
197.(c)	198.(d)	199.(a)	200.(a)
201.(c)	202.(a)	203.(c)	204.(a)



205.(b)	206.(c)	207.(b)	208.(c)
209.(d)	210.(a)	211.(d)	212.(c)
213.(b)	214.(a)	215.(b)	216.(d)
217.(b)	218.(b)	219.(c)	220.(a)
221.(a)	222.(c)	223.(c)	224.(d)
225.(a)	226.(b)	227.(c)	228.(b)
229.(a)	230.(c)	231.(b)	232.(b)
233.(d)	234.(c)	235.(c)	236.(b)
237.(c)	238.(d)	239.(b)	240.(c)
241.(d)	242.(c)	243.(d)	244.(b)
245.(a)	246.(a)	247.(a)	248.(b)
249.(c)	250.(d)	251.(c)	252.(c)
253.(a)	254.(a)	255.(c)	256.(b)
257.(a)	258.(b)	259.(a)	260.(b)
261.(d)	262.(a)	263.(a)	264.(d)
265.(b)	266.(b)	267.(a)	268.(a)
269.(d)	270.(d)	271.(a)	272.(d)
273.(d)	274.(b)	275.(d)	276.(a)
277.(d)	278.(a)	279.(a)	280.(c)
281.(d)	282.(d)	283.(b)	284.(c)
285.(b)	286.(c)		

Solutions :-

Sol.1.(b) Given :- $\frac{\sqrt{5+x} + \sqrt{5-x}}{\sqrt{5+x} - \sqrt{5-x}} = 7$

By using componendo and Dividendo

$$\frac{\sqrt{5+x} + \sqrt{5-x}}{\sqrt{5+x} - \sqrt{5-x}} = \frac{7}{1}$$

$$\frac{\sqrt{5+x} + \sqrt{5-x} + \sqrt{5+x} - \sqrt{5-x}}{\sqrt{5+x} + \sqrt{5-x} - \sqrt{5+x} + \sqrt{5-x}}$$

$$= \frac{7+1}{7-1} \Rightarrow \frac{\sqrt{5+x}}{\sqrt{5-x}} = \frac{8}{6} = \frac{4}{3}$$

Square on both sides

$$\frac{5+x}{5-x} = \frac{16}{9}$$

$$45 + 9x = 80 - 16x$$

$$25x = 35 \Rightarrow x = \frac{35}{25} = \frac{7}{5}$$

Sol.2.(c)

Let the number of trees in each row be n

$$\text{Then } n^2 = 4112 - 16 = 4096$$

$$n = \sqrt{4096} = 64$$

Sol.3.(b) A/Q,

Let smaller number = x and larger = $(7x + 14)$

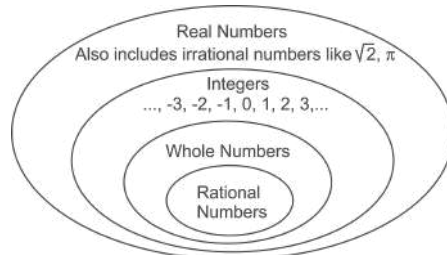
$$(7x + 14) - x = 1370$$

$$6x = 1356 \Rightarrow x = 226$$

Sol.4.(a)

2	187	1
2	93	1
2	46	0
2	23	1
2	11	1
2	5	1
2	2	0
2	1	1
2	0	

$$(187)_{10} = (10111011)_2$$

Sol.5.(c) Real numbers

Sol.6.(b) For any two consecutive natural numbers n and $(n+1)$, there are $2n$ non-square numbers between their squares.

$$n = 30$$

$$\text{Next number} = (n+1) = 31$$

$$\text{Total non-square numbers between } 30^2 \text{ and } 31^2 = 2 \times 30 = 60$$

Sol.7.(b) (i) $90 = 90 \times 90 = 8100$

(ii) $600 = 600 \times 600 = 360000$

(i) has 2 zeros and (ii) has 4 zeros.

Sol.8.(b) $x + y = \frac{5}{3} + \sqrt{5} = \frac{5 + 3\sqrt{5}}{3}$

Their sum will also be an irrational number

Sol.9.(c) 45yz0

For divisibility of 40 number must be divisible by 8 and 5

Divisibility of 5:- A number is divisible by 5 if its last digit is 0 or 5

Divisibility of 8:- if the last three digits of the number are either all zeros or are divisible by 8:

Maximum possible value of y and z be 8 and 8 respectively,

$$\text{So, the value of } (y + z) = 8 + 8 = 16$$

Sol.10.(c) Let number of chairs = x

When one person is seated on each chair then

$$\text{Total number of men} = x + 1 \dots (i)$$

When two person is seated on each chair then

$$\text{Total number of men} = 2x - 2 \dots (ii)$$

From (i) and (ii)

$$x + 1 = 2x - 2 \Rightarrow x = 3$$

$$\text{Hence number of chairs} = 3$$

Sol.11.(d) According to question

$$15A + 12O = 447 \dots (i)$$

For inconsistent and no possible solutions $\frac{a1}{a2} = \frac{b1}{b2} \neq \frac{c1}{c2}$

By checking options one by one we find that only option (d) satisfies the condition.

In option (d)

$$35A + 28O = 1029 \dots (ii)$$

From (i) and (ii)

$$\frac{a1}{a2} = \frac{15}{35} = \frac{3}{7}$$

$$\frac{b1}{b2} = \frac{12}{28} = \frac{3}{7}$$

$$\frac{c1}{c2} = \frac{447}{1029} = \frac{149}{343}$$

$$\text{hence, } \frac{a1}{a2} = \frac{b1}{b2} \neq \frac{c1}{c2}$$

Sol.12.(a) Let the number of notes ₹20 and ₹10 be x and y respectively,

According to the question,

$$20x + 10y = ₹ 390 \dots (i)$$

$$10x + 20y = ₹ 300 \dots (ii)$$

Add eq. (i) and (ii)

$$30x + 30y = 690$$

$$x + y = 23 \Rightarrow y = 23 - x$$

Put the value of y in eq. (i)

$$20x + 230 - 10x = 390$$

$$10x = 160 \Rightarrow x = 16$$

Sol.13.(d) Let cost of 1 biscuit, 1 chocolate, 1 cup coffee be x, y, z respectively.

According to question,

$$2x + y = 69 \dots (i)$$

$$2y + 3z = 127 \dots (ii)$$

$$3x + 4y + 2z = 229 \dots (iii)$$

Adding (i) and (iii)

$$5x + 5y + 2z = 69 + 229$$

$$5x + 5y + 2z = 298 \dots (iv)$$

now, Multiply eq. (i) by 2 and then subtracting eq. (ii) we get

$$4x - 3z = 138 - 127$$

$$4x - 3z = 11 \dots (v)$$

Multiply eq. (ii) by 2 and then subtracting eq. (iii) we get

$$-3x + 4z = 254 - 229$$

$$-3x + 4z = 25 \dots (vi)$$

By solving eq. (v) and (vi) we get

$$z = 19$$

From eq. (iv) adding $3z$ we get

$$5x + 5y + 5z = 298 + 19 \times 3$$

$$= 298 + 57 = 355$$

Sol.14.(b) Let the weight of apples and oranges is 'a' and 'b' respectively.

According to the question,

$$8a + 10b = 5 \dots (i)$$

$$12a + 20b = 9 \dots (ii)$$

Multiply by 2 in e.q. (i) then subtract e.q. (ii) from (i) we get, $4a = 1$

$$a = \frac{1}{4} \text{ (put the value in e.q. (i) we get,}$$

$$8 \times \frac{1}{4} + 10b = 5 \Rightarrow 10b = 3 \Rightarrow b = \frac{3}{10}$$



Weight of 15 apples and 24 oranges

$$= 15 \times \frac{1}{4} + 24 \times \frac{3}{10} = 3.75 + 7.2$$

$$= 10.95 \text{ kg (approx 11 kg)}$$

Sol.15.(c) Let winner = A,
1st runner up = B, 2nd runner up = C
and nota = N

According to question

$$A = 2C, B = N + 900 \text{ and } C$$

$$= N + 900 - 150$$

Now,

$$A + B + C = 6N$$

$$2C + B + C = 6N$$

$$3C + B = 6N$$

$$3C + N + 900 = 6N$$

$$3(N + 900 - 150) + N + 900 = 6N$$

$$3N + 2250 + N + 900 = 6N$$

$$3150 = 2N \Rightarrow N = 1575$$

$$\text{Hence, } B = N + 900 = 2475$$

$$C = N + 900 - 150 = 2325 \Rightarrow A = 2C = 4650$$

$$\text{Total votes} = A + B + C + N$$

$$= 4650 + 2475 + 2325 + 1575 = 11025$$

Sol.16.(a) Factor of 14400 = $2^6 \times 3^2 \times 5^2$

$$= 2^6 \times 3^2 \times 5^2 = 18 \times (2^5 \times 5^2)$$

Divisible by 18 then remaining factor

$$= 6 \times 3 = 18$$

$$= 2^6 \times 3^2 \times 5^2 = 36 \times (2^4 \times 5^2)$$

Divisible by 36 then remaining factor

$$= 5 \times 3 = 15$$

Required factor divisible by 18 but not by

$$36 = 18 - 15 = 3$$

Sol.17.(d) According to question,

Amount of Raju after 6 weeks

$$= 11000 + 7 \times 5000 = ₹ 46000$$

Now,

Amount of Ramesh after 6 weeks

$$= ₹ 60000 - 7 \times 2000 = ₹ 46000$$

Sol.18.(a) Divisibility of 11: The difference of the sum of digits at odd position and sum of digits at even position in a number is 0 or divisible by 11.

$$\text{Now, } (3 + 2 + 3 + k) - (c + 9 + 3) = 0$$

$$\Rightarrow (8 + k) = (c + 12) \dots\dots(i)$$

Divisibility of 5 : If the last digit of the number should be 0 or 5 then that number will be divisible by 5.

Value of k

$$= 5 \text{ (given we take only natural no.)}$$

Put value of k in equation (i), we get

$$\Rightarrow (8 + 5) = (c + 12) \Rightarrow c = 13 - 12 = 1$$

So,

$$c + k = 1 + 5 = 6$$

Sol.19.(d) Let the numbers be a and b and the divisor be d. The remainders when a and b are divided by d are 6 and 7 respectively. Thus,

$$a = dq_1 + 6, b = dq_2 + 7$$

For their sum

$$a + b = d(q_1 + q_2) + 13 \dots (i)$$

When a + b is divided by d, the remainder is 5,

$$a + b = dq + 5 \dots (ii)$$

Equating (i) and (ii) we get,

$$d(q_1 + q_2) + 13 = dq + 5$$

$$d(q_1 + q_2 - q) = -8$$

Thus, d must divide 8. Testing values, we find that d = 8 satisfies the condition.

Therefore, the divisor is 8

Sol.20.(c)

1738b9c is divisible by 12 (i.e. 4×3)

Now, **divisibility of 3** :- sum of digits of number be divisible by 3

$$1738b9c = 1 + 7 + 3 + 8 + b + 9 + c$$

$$= 28 + b + c$$

So, possible values of (b + c)

$$= 2, 5, 8, 11, 14 \text{ etc}$$

Now, **Divisibility of 4** :- last two digit of number be divisible by 4 $\Rightarrow$ number = 9c

Possible values of c = 2 and 6

So, for least value of (c - b)

We should take c = 6 and b = 5

$$\{b + c = 5 + 6 = 11\}$$

Hence, required value of (c - b) = 6 - 5 = 1

Sol.21.(d)

$$\begin{array}{r} 7 \overline{) 36952} \\ \underline{235} \\ 19 \\ \underline{14} \\ 5 \end{array} \quad \begin{array}{r} 5 \overline{) 5210} \\ \underline{5} \\ 2 \end{array}$$

$$\begin{array}{r} 3 \overline{) 103} \\ \underline{9} \\ 1 \end{array} \quad \begin{array}{r} 2 \overline{) 321} \\ \underline{2} \\ 1 \end{array}$$

Hence, sum of remainder

$$= 5 + 2 + 1 + 1 = 9$$

Sol.22.(b) Divisibility rule of 3 :

For any no to be divisible by 3, the sum of its digits should be divisible by 3.

$$57 = 19 \times 3$$

Given that, 468312 is divisible by (19×3)

On arranging it in descending order, digits remain the same. so it will always be divisible by 3.

Sol.23.(a) If the number is divisible by 5 means the last digit is either 0 or 5.

$$\text{Let } Q = 0$$

Now we have to find the value of P.

For divisibility by 3, the sum of the digits should be divisible by 3.

$$\Rightarrow P = 2, 5, 8$$

But the numbers 3420, 3450 and 3480 are not divisible by 7.

Now, take the value of Q = 5.

For divisibility by 3, we get,

$$P = 3, 6, 9$$

But by checking for divisibility by 7, we get P = 6.

$$\text{Hence, } P + Q = 6 + 5 = 11.$$

Sol.24.(d) Let six digit number = $33 \times x$

$$\text{New number} = 33 \times x + 21$$

$$= 3\{(11 \times x) + 7\}$$

We can say that new number is divisible by 3

Sol.25.(a) $24 = 8 \times 3$

For 11p9q4 to be divisible by 8, its last 3 digits i.e. 9q4 should be divisible by 8

Then, the possible values of q are 4, 8.

But for the greatest value, q should be 8.

For 11p984 to be divisible by 3, the sum of its digits i.e. $1 + 1 + p + 9 + 8 + 4 = 23 + p$ should be divisible by 3.

For greatest value, p should be 7

$$\text{Hence, } pq = 7 \times 8 = 56$$

Sol.26.(d) Given :- $\frac{9 + 9^2 + \dots + 9^{(2n+1)}}{6}$

$$\frac{9}{6} = \text{Rem } 3, \frac{9^2}{6} = \text{Rem } 3, \dots, \frac{9^{2n+1}}{6}$$

$$= \text{Rem } 3$$

$$\text{So, } \frac{3 \times (2n+1)}{6} = \frac{6n+3}{6} = \text{Rem } 3$$

Sol.27.(d) Remainder when sum of these numbers are divided by same divisor

$$= 57 + 57 = 114$$

According to the question,

$$\frac{114}{\text{divisor}} = \text{Rem.}(49)$$

$$\text{Required divisor} = (114) - 49 = 65$$

Sol.28.(d)

$$\text{Divisor} = 5 \times \text{Remainder} = 5 \times 44 = 220$$

$$\text{Quotient} = \frac{220}{11} = 20$$

$$\text{Dividend} = 220 \times 20 + 44 = 4444$$

Sol.29.(d)

Co-prime factor of 88 = (8 and 11)

Least value of y, when $y24$ divisible by

$$8 :- y = 0$$

Now, checking divisibility of 11

$$\Rightarrow (7 + 0 + 5 + 3 + 2) - (8 + x + 3 + 0 + 4) = 0$$

$$\Rightarrow 17 - (15 + x) = 0 \Rightarrow 2 - x = 0 \Rightarrow x = 2$$

$$\text{Now, value of } (x + y) = 2 + 0 = 2$$

Sol.30.(a) We know that:-

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

$$\text{Ratio} \rightarrow \text{initial} : \text{final}$$

$$\text{Dividend} \rightarrow 10 : 9$$

$$\text{Divisor} \rightarrow 5 : 4$$

$$\text{Initial quotient} = 24$$

$$\text{Initial number} = 24 \times 5 = 120$$

$$120 \text{ which is equal to } 10x. \Rightarrow x = 12$$

$$\text{Final dividend } (9x) = 108$$

$$\text{Final quotient} = \frac{108}{4} = 27$$

Sol.31.(d)

$$\text{Ratio :- initial : final}$$

$$\text{Divisor :- } 4 : 5$$

$$\text{Dividend :- } 10 : 11$$

$$\text{Initial quotient} = 25$$

$$\text{Initial dividend } (10 \text{ units}) = 25 \times 4 = 100$$

$$\text{Then final dividend } (11 \text{ units}) = 110$$



$$\text{Final quotient} = \frac{110}{5} = 22$$

Sol.32.(a) Divisibility of 33 : the given number must be divisible by 11 and 3.

Divisibility of 11 : the difference between the sum of odd place digit and the sum of even place digit is either 0 or 11.

Divisibility of 3 : the sum of all digits of the given number must be divisible by 3.

The given number = $7x1yyx$

$$(7 + 1 + y) - (x + y + x) = 0 \text{ or } 11$$

$$8 - 2x = 0 \Rightarrow x = 4$$

$$\text{Now, } (7 + 4 + 1 + y + y + 4) = 16 + 2y$$

Putting value of $y = 1$

$$16 + 2y = 18 \text{ [divisible by 3]}$$

$$\text{So, required sum} = 4 + 1 = 5$$

Sol.33.(b) According to the question,

$$10 \times 12 = (10 + 5) \times x$$

Number of tree in each row(x)

$$= \frac{12 \times 10}{15} = 8$$

$$\text{Sol.34.(d)} \quad 720 = 2^4 \times 3^2 \times 5^1$$

$$\text{Total no of factors} = 5 \times 3 \times 2 = 30$$

$$\text{Removing 1 and } 720 = 30 - 2 = 28$$

$$\text{Sol.35.(b)} \quad (14)^{\frac{1}{3}} = (14)^{\frac{4}{12}}, (12)^{\frac{1}{2}}$$

$$= (12)^{\frac{6}{12}}, (16)^{\frac{1}{6}} = (16)^{\frac{2}{12}} \& (25)^{\frac{1}{12}}$$

$$= (25)^{\frac{1}{12}}$$

Therefore, $(25)^{\frac{1}{12}}$ is smallest fraction

Sol.36.(c)

series = -9, -6, -3, ... (series is in the A.P.)

So, $a = -9$, $d = 3$

Sum of n number

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$\Rightarrow 45 = \frac{n}{2}[-18 + (n-1)3]$$

$$\Rightarrow 3n^2 - 21n - 90 = 0 \Rightarrow n = 10 \text{ and } -3$$

(no of terms never be -ve)

Therefore, no. of terms in the given series = 10 term

Sol.37.(d)

$$\text{I. } (100^2 - 99^2) + (98^2 - 97^2) + (96^2 - 95^2)$$

$$+ (94^2 - 93^2) + \dots + 22^2 - 21^2 = 4840$$

$$= (100 + 99)(100 - 99) + (98 - 97)(98 + 97) + (96 - 95)(96 + 95) + \dots + (22 - 21)(22 + 21)$$

$$= (100 + 99 + 98 + 97 + 96 + 95 + \dots + 21)$$

$$= \frac{1}{2} \times 80 \times (21 + 100) = 40 \times 121 = 4840$$

So, LHS = RHS

$$\text{II. } (k^2 + \frac{1}{K^2})(K - \frac{1}{K})(K^4 + \frac{1}{K^4})(K + \frac{1}{K})$$

$$(K^4 - \frac{1}{K^4}) = K^{16} - \frac{1}{K^{16}}$$

Taking L.H.S

$$= (k^2 + \frac{1}{K^2})(k^2 - \frac{1}{K^2})(K^4 + \frac{1}{K^4})(K^4 - \frac{1}{K^4})$$

$$= (K^4 - \frac{1}{K^4})(K^4 + \frac{1}{K^4})(K^4 - \frac{1}{K^4})$$

$$= (K^8 - \frac{1}{K^8})(K^4 - \frac{1}{K^4})$$

So, LHS $\neq$ RHS

Clearly, we can see that only statement 1 is correct.

Sol.38.(b) $33 = 11 \times 3$

For 52A6B7C to be divisible by 11, the

difference of the sum of its alternate

digits i.e. $(5 + A + B + C) - (2 + 6 + 7)$

$= A + B + C - 10$, should be divisible by 11.

For this, we have: $A + B + C - 10 = 0/11 \dots$

$$\Rightarrow A + B + C = 10$$

As we know, the sum of three odd no's are always odd. It suggests that there should be 1 even prime no i.e.2. So, we get A, B, C

as 2, 3, 5

For 52A6B7C to be divisible by 3, the sum of its digits i.e.

$$5 + 2 + A + 6 + B + 7 + C = 20 + A + B + C, \text{ should be divisible by 3.}$$

Now, putting $A + B + C = 10$,

we get : $20 + 10 = 30$, divisible by 3.

For maximum value of

$$2A + 3B + C; A = 3, B = 5, \text{ and } C = 2$$

$$\text{So, } 2A + 3B + C = 2 \times 3 + 3 \times 5 + 2$$

$$= 6 + 15 + 2 = 23$$

Sol.39.(c) Jay answered 4 correct answers and + 5 marks are given for every correct answer that means he scored 20 marks for his correct answer But Jay scored - 12 marks that means he loses his 20 marks of correct answers also.

Total marks deducted due to incorrect answers = $20 + 12 = 32$

For every incorrect answer 2 marks are deducted.

$$\text{So, no of incorrect answers} = \frac{32}{2} = 16$$

Sol.40.(b) After arranging the digits in descending order and subtracting 65 from the new no, we get :

$$853210 \rightarrow 853210 - 65 = 853145, \text{ which is divisible by 5 } (\because \text{last digit is 5})$$

Sol.41.(a)

$$S_1 = 2, 9, 16, \mathbf{23}, 30, 37, 44, \mathbf{51}, \dots, 632$$

$$S_2 = 7, 11, 15, 19, \mathbf{23}, 27, 31, 35, 39, 43, 47, \mathbf{51},$$

Common terms between given series

$$= 23, 51, \dots$$

$$\text{Common difference}(d) = 51 - 23 = 28$$

ATQ,

$$23 + (n-1)28 \leq 632$$

$$(n-1)28 \leq 609 \Rightarrow n-1 \leq 21.75$$

$$n \leq 22.75 \Rightarrow \text{so, } n = 22$$

$$\text{Now, } S = \frac{22}{2} [2 \times 23 + (22 - 1)28]$$

$$S = 11[46 + 21 \times 28]$$

$$S = 11[46 + 588] = 11 \times 634 = 6974$$

Sol.42.(a)

As prime numbers have 2 factors

Only the squares of prime numbers will

have three factors. $\Rightarrow$ Let $r = 9$ and $s = 4$

And $q = 49$ and $p = 25$

$$\frac{p-q-1}{r-s} = \frac{25-49-1}{9-4} = \frac{-25}{5} = -5$$

Putting $s = 4$ in all options we find only option (a) satisfies this value.

$$\text{Sol.43.(c)} \quad 0.0625 = \frac{1}{16},$$

$$\text{So, } y = \frac{9}{16} - \frac{1}{16} = \frac{8}{16} = \frac{1}{2}$$

$$\text{Again, } x + y + z = 2\frac{3}{12} = \frac{27}{12}$$

$$x + z = \frac{27}{12} - \frac{1}{2} = \frac{27-6}{12} = \frac{7}{4}$$

$$\text{Sol.44.(b)} \quad (30)^5 \times (24)^5$$

$$= (2 \times 3 \times 5)^5 \times (2^3 \times 3)^5$$

$$= 2^{20} \times 3^{10} \times 5^5$$

Number of prime factors = $(20 + 10 + 5)$

Number of prime factors = 35

Sol.45.(b) HINT : ab and ba both can be prime only and only when both are odd number

Such numbers are 13, 31, 17, 71, 37, 73, 79, 97

$$\text{Sum} = 13 + 31 + 17 + 71 + 37 + 73 + 79 + 97 = 418$$

Sol.46.(d) The whole number which divides $2^{20} + 1$ also divides the multiple of $2^{20} + 1 \Rightarrow 2^{60} + 1 = (2^{20} + 1)(2^{40} + 1 - 2^{20})$

Using formula :

$$a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$$

$$2^{60} + 1 \text{ is the multiple of } 2^{20} + 1$$

so, $2^{60} + 1$ is completely divisible by that whole number.

Sol.47.(a)

$$7^{81} + 7^{82} + 7^{83} = 7^{81} \times (1 + 7 + 7^2) = 7^{81}$$

$$\times (1 + 7 + 49) = 7^{81} \times (57) = 7^{80}$$

$$\times (7 \times 57) = 7^{80} \times (399)$$

Option (a) follows.

Sol.48.(c) Let N be the number which gives Q as quotient and 15 as remainder when divided by d .

Thus, $d > 15$

$$N = d \times Q + 15$$

$$10N = 10(d \times Q) + 144 + 6$$

clearly d is a multiple of 144 which are: 2, 3, 4, 6, 8, 9, 12, 16, 18, 24 and so on.

The least possible value of d is 16.

($d > 15$)



Sol.49.(c) When 200 is divided by x , the remainder is 8. So, the number exactly divisible by x is 192.

Factors of 192 = 1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 64, 96, 192

The remainder is always less than the divisor,

so $x > 8$

$\Rightarrow$ Desired values are 12, 16, 24, 32, 48, 64, 96, 192.

Sol.50.(a) Total no's less than 1000 which is divisible by 5 = $\frac{1000}{5} - 1$
 $= 200 - 1 = 199$

Total no's less than 1000 which is divisible by 7 = $\frac{1000}{7} = 142$

Total no's less than 1000 which is divisible by 35 = $\frac{1000}{35} = 28$

Required no = $199 + 142 - (2 \times 28)$
 $= 285$

Sol.51.(b) Now, we know that the number that is divisible by 7, 11 and 13 is 1001 if 1001 is multiplied with a three digit number say, 'abc'

then the product comes out to be abcabc
 $\Rightarrow x = 4, y = 7, z = 9$

Putting these values in the equation
 $\{(y + z) \div x\}$
 $= \{(7 + 9) \div 4\} = 16 \div 4 = 4$

Sol.52.(a) As per the options the difference between all the two values of the given four options is '2'

Thus, in such cases : if there is the same difference between the numbers then, The greatest value with the square root of the given options fetches the smallest value. So $\sqrt{401} - \sqrt{399}$ will give the smallest value.

Sol.53.(d) Let the numbers be x and y .

According to the question

$x - y = 2001$ (1)

And $9y + 41 = x$ (2)

Put the value of x in eq (1)

$9y + 41 - y = 2001 \Rightarrow 8y = 1960$

$y = 245 \Rightarrow x = 9(245) + 41 = 2246$

Required sum = $2 + 2 + 4 + 6 = 14$

Sol.54.(c) Let the number = $10x + y$

According to the question

$(10x + y)(x + y) = 424$ (1)

And

$(10y + x)(x + y) = 280$ (2)

Divide equation (1) by (2)

$\Rightarrow \frac{(10x + y)}{(10y + x)} = \frac{424}{280} \Rightarrow \frac{(10x + y)}{(10y + x)} = \frac{53}{35}$

$\Rightarrow 350x + 35y = 530y + 53x$

$\Rightarrow 297x = 495y \Rightarrow \frac{x}{y} = \frac{5}{3}$

Put this value in any of the equations

$(10x + y)(x + y) = [10(5) + 3](5 + 3) = 424$

Or

$(10y + x)(x + y) = [10(3) + 5](5 + 3) = 280$

Clearly 5 and 3 are the desired values and the sum of the digits = $5 + 3 = 8$

Sol.55.(d) Given,

$x = (633)^{24} - (277)^{38} + (266)^{54}$

Unit digit of $(633)^{24} = 3^4 = 1$

Unit digit of $(277)^{38} = 7^2 = 9$

Unit digit of $(266)^{54} = 6^2 = 6$

Unit digit of $x = 1 - 9 + 6 = -2$

But unit digit can't be negative so, required unit digit = $10 + (-2) = 8$

Sol.56.(d) Let the number = $10x + y$

According to the question

$(x + y) = \frac{1}{7}(10x + y)$ (1)

And, $y = x - 4$

Put this value in equation (1)

$(x + x - 4) = \frac{1}{7}(10x + x - 4)$

$14x - 28 = 11x - 4$

$x = 8$ and $y = 8 - 4 = 4$

the number obtained on reversing the digit = $10y + x = 10(4) + 8 = 48$

Required remainder = $\frac{48}{7} = 6$

Sol.57.(b) 44

	44
4	1936
+ 4	-16
84	336
	-336
	0

Sol.58.(d) Difference between the largest and the smallest number

$9652 - 2569 = 7083$

Sol.59.(c) Since, $\sqrt{1354.24} = 36.8$

Sol.60.(d) $22.\overline{4} + 11.5\overline{67} - 33.\overline{59}$

$\Rightarrow 22.444444 \dots \infty + 11.5676767 \dots \infty -$

$33.599999 \dots \infty$

$= 0.412121212 \dots \infty = 0.4\overline{12}$

Sol.61.(b) $\frac{6}{11} \approx 0.5455, \frac{5}{8} = 0.625,$

$\frac{4}{7} \approx 0.5714$ and $\frac{3}{5} = 0.6$

Ascending order = $\frac{6}{11} < \frac{4}{7} < \frac{3}{5} < \frac{5}{8}$

Sol.62.(c) Given :-

$x = \sqrt{3} + \sqrt{5} = 1.732 + 2.236 = 3.968$

$y = \sqrt{8} + \sqrt{3} = 2.828 + 1.732 = 4.56$

so, $x < y$

Sol.63.(a)

$x = \sqrt{2} + \sqrt{3} = 1.41 + 1.73 = 3.14$

$y = \sqrt{6} + \sqrt{1} = 2.44 + 1 = 3.44$

$z = 3\sqrt{2} + 1 = 3 \times 1.41 + 1 = 5.23$

So the order is: $x < y < z$

Sol.64.(a) Given:-

$(\frac{1}{4 - \sqrt{15}}) - (\frac{1}{\sqrt{15} - \sqrt{14}}) + (\frac{1}{\sqrt{14} - \sqrt{13}})$

$- (\frac{1}{\sqrt{13} - \sqrt{12}}) + (\frac{1}{\sqrt{12} - \sqrt{11}}) -$

$(\frac{1}{\sqrt{11} - \sqrt{10}}) + (\frac{1}{\sqrt{10} - 3})$

Using Rationalization Method

$x = (\frac{1}{4 - \sqrt{15}} \times \frac{4 + \sqrt{15}}{4 + \sqrt{15}}) - (\frac{1}{\sqrt{15} - \sqrt{14}} \times$

$\frac{\sqrt{15} + \sqrt{14}}{\sqrt{15} + \sqrt{14}}) + (\frac{1}{\sqrt{14} - \sqrt{13}} \times \frac{\sqrt{14} + \sqrt{13}}{\sqrt{14} + \sqrt{13}})$

$- (\frac{1}{\sqrt{13} - \sqrt{12}} \times \frac{\sqrt{13} + \sqrt{12}}{\sqrt{13} + \sqrt{12}}) +$

$(\frac{1}{\sqrt{12} - \sqrt{11}} \times \frac{\sqrt{12} + \sqrt{11}}{\sqrt{12} + \sqrt{11}}) - (\frac{1}{\sqrt{11} - \sqrt{10}} \times$

$\frac{\sqrt{11} + \sqrt{10}}{\sqrt{11} + \sqrt{10}}) + (\frac{1}{\sqrt{10} - 3} \times \frac{\sqrt{10} + 3}{\sqrt{10} + 3})$

$\frac{4 + \sqrt{15} - \sqrt{15} - \sqrt{14} + \dots - \sqrt{11} - \sqrt{10} + \sqrt{10} + 3}{1}$

$= 4 + 3 = 7$

Then, $x + 1 = 7 + 1 = 8$

Sol.65.(c)

$a = \sqrt{5} - \sqrt{3} \Rightarrow a = 2.236 - 1.732 = 0.504$

$b = \sqrt{3} - \sqrt{2} \Rightarrow b = 1.732 - 1.414 = 0.318$

Hence, $a > b$

Sol.66.(d) On checking all options one by one, only option (d) satisfies,

$= (\sqrt{8} + \sqrt{6})^2 - 2\sqrt{48}$

$= 8 + 6 + 2\sqrt{48} - 2\sqrt{48} = 14$

Sol.67.(a) $y = 4x + 2$ (i)

$y = -2x + 8$ (ii)

From eq. (i) and (ii),

$\Rightarrow 4x + 2 = -2x + 8$

$\Rightarrow 6x = 6 \Rightarrow x = 1$

Put the value x in eq. (i), we get

$\Rightarrow y = 6$

So, the value of $(x, y) = (1, 6)$

Sol.68.(a) Given :- $0.232323 \dots$

$\Rightarrow 0.232323 \dots = 0.\overline{23} = \frac{23}{99}$

Sol.69.(b) $0.48 \times 0.7 = 0.336$

Sol.70.(c) $(\sqrt{5} + \sqrt{6})^2 = 5 + 6 + 2\sqrt{30}$
 $= 11 + 2\sqrt{30}$

$(\sqrt{7} + \sqrt{4})^2 = 7 + 4 + 2\sqrt{28}$

$= 11 + 2\sqrt{28}$

$\sqrt{5} + \sqrt{6} > \sqrt{7} + \sqrt{4}$

Incorrect relation in (i) and (ii)

Sol.71.(a) $(\sqrt{11} + \sqrt{3})^2$

$= 11 + 3 + 2\sqrt{11 \times 3} = 14 + 2\sqrt{33}$

$(\sqrt{12} + \sqrt{2})^2 = 12 + 2 + 2\sqrt{12 \times 2}$

$= 14 + 2\sqrt{24}$



Hence, $\sqrt{11} + \sqrt{3} > \sqrt{12} + \sqrt{2}$

Sol.72.(d) The unit digit of a not perfect square number always (2, 3, 7, 8)

Hence, 100788 is a not perfect square number

Sol.73.(d) Let number = x

ATQ,

$$13(2796 - x^2) = 35711$$

$$(2796 - x^2) = \frac{35711}{13} = 2747$$

$$x^2 = 2796 - 2747 = 49 \Rightarrow x = \pm 7$$

Sol.74.(b) $\sqrt{8} = 2.82$, $\sqrt{11} = 3.31$

$$\sqrt{15} = 3.87, \sqrt{17} = 4.12$$

$$\sqrt{9} < 3.1 \Rightarrow 3 < 3.1 \text{ (true)}$$

$$\sqrt{15} < \sqrt{17} \Rightarrow 3.87 < 4.12 \text{ (true)}$$

$$\sqrt{8} < \sqrt{11} \Rightarrow 2.82 < 3.31 \text{ (true)}$$

Sol.75.(b) When any rational number is added to an irrational number, the result is always irrational.

$$\sqrt{2} + 7 = 1.414..... + 7 = 8.414 \text{ (irrational)}$$

$$\text{or } \sqrt{2} + 0 = \sqrt{2} \text{ (irrational)}$$

Sol.76.(a) $y = 7x - 4$ (i)

$y = -2x + 14$ (ii)

According to question

$$7x - 4 = -2x + 14$$

$$7x + 2x = 14 + 4$$

$$9x = 18 \Rightarrow x = 2$$

Substituting in equation (i)

$$y = 7x - 4$$

$$y = 7 \times 2 - 4 = 14 - 4 = 10$$

$$(x, y) = (2, 10)$$

Sol.77.(b) Whole number = A whole number is a number that is not negative and has not fractions and decimals. The correct answer is 1.

Sol.78.(c) Given :- $\frac{\sqrt{8+x} + \sqrt{8-x}}{\sqrt{8+x} - \sqrt{8-x}} = 5$

$$\text{Let, } a = \sqrt{8+x} \text{ and } b = \sqrt{8-x}$$

$$\Rightarrow \frac{a+b}{a-b} = 5$$

$$\Rightarrow a+b = 5a-5b$$

$$\Rightarrow -4a = -6b \Rightarrow \frac{a}{b} = \frac{3}{2}$$

$$\text{Now, } \frac{\sqrt{8+x}}{\sqrt{8-x}} = \frac{3}{2}$$

Square on both sides

$$\frac{8+x}{8-x} = \frac{9}{4} \Rightarrow 32+4x = 72-9x$$

$$-40 = -13x \Rightarrow x = \frac{40}{13}$$

Sol.79.(b) 0' is an integer number. but is not a natural number

Sol.80.(b) After going through all the options, option (b) satisfies.

Square root of 16: The square root of 16 is 4. This is an integer.

Sol.81.(c) 25 needs to be added to 5600 to make it a perfect square, because $74^2 = 5476$ and $75^2 = 5625$.

Sol.82.(c) Whole numbers = {0, 1, 2, 3, ...}

Integers = {..., -3, -2, -1, 0, 1, 2, ...}

The number that is an integer but not a whole number is -5.

Sol.83.(c) Real number = $\frac{3}{5}$, 7 and -3

Square root of 5 is an irrational number not a real number.

Sol.84.(a) According to question, 1 cubic meter = 1000L

$$\text{Number of buckets} = \frac{1000 \text{ l}}{12.5 \text{ l}} = 80$$

Sol.85.(d) decimal 0.785 is best classified as a rational number. It is also a terminating decimal.

Sol.86.(b) $\frac{7}{5}$ is a Rational Number and a Terminating Decimal.

Sol.87.(d) According to question, Total number of servings

$$= \frac{15}{0.85} \times 4.25 = 75$$

Sol.88.(b) let x be number

According to question,

$$\Rightarrow x^2 - 5x = 150 \Rightarrow x(x-5) = 150$$

By options, put value x = 15

$$= 15(15-5) = 15 \times 10 = 150$$

Sol.89.(b)

Both Positive and Negative Numbers

Sol.90.(a) As Integers are whole numbers {..., -2, -1, 0, 1, 2, ...}

Rational numbers are numbers that can be expressed as a fraction $\frac{p}{q}$, where $q \neq 0$

Therefore, $\sqrt{3}$ cannot be expressed as a fraction and has a non-terminating, non-repeating decimal expansion.

Sol.91.(b) Whole numbers include all positive integers and zero.

Sol.92.(d) Integers are whole numbers that can be positive, negative, or zero, with no fractional or decimal components.

Sol.93.(d) Non-zero positive numbers are the numbers that are greater than zero and do not include zero (0).

For example – Natural Numbers

(1, 2, 3, 4, 5, ...)

Sol.94.(c)

Squares between 729 and 784 are $27^2 = 729$ and $28^2 = 784$

Lets the number be = x

$$\text{Then, } x + 100 = n^2 \text{ and } 729 < x < 784$$

put, $n = 29$

$$x + 100 = 29^2 = 841 \Rightarrow x = 741$$

Sol.95.(b) 0' is an integer number. but is not a natural number

Sol.96.(c) All rational and irrational numbers together form the Real numbers

Sol.97.(a) $\frac{147}{22}$

$$\text{Mixed number} = 6\frac{15}{22}$$

Sol.98.(c) Largest number = $15^2 = 225$

Smallest number $13^2 = 169$

$$\text{Total numbers} = 225 - 169 - 1 = 55$$

$$\text{Sol.99.(b)} \frac{7\sqrt{5} \times \sqrt{20}}{\sqrt{4}} = \frac{7\sqrt{100}}{2}$$

$$= \frac{7 \times 10}{2} = \frac{70}{2} = 35$$

$\therefore 35$ is a rational number

Sol.100.(c) All irrational numbers (numbers that cannot be expressed as a fraction ($\frac{p}{q}$)) are part of the real numbers set.

$$\text{Sol.101.(b)} \sqrt{196} = 14$$

Sol.102.(a) Every natural number is always an integer.

Example :- 5 is a natural number and also an integer.

$$\text{Sol.103.(d)} (73)^2 = 5329$$

Sol.104.(d) Both $\sqrt{6}$ and $\sqrt{10}$ are irrational numbers, and their sum $\sqrt{6} + \sqrt{10}$ is also irrational because it cannot be expressed as a fraction $\frac{p}{q}$.

Sol.105.(d) Irrational

$$\text{Sol.106.(b)} 4.75 = \frac{475}{100} = \frac{19}{4} = 4\frac{3}{4}$$

Sol.107.(c) In this type of questions, We check only unit digit

Square of 7 = 49

So, the unit digit of the square of 67 is 9.

Sol.108.(a) Integers are numbers that can be represented on the number line, and all integers belong to the set of real numbers.

Therefore, statement (a) **All integers are real** is correct.

Sol.109.(a)



All natural numbers are whole numbers

Explanation: Natural numbers are: 1, 2, 3, 4, ...

Whole numbers are: 0, 1, 2, 3, 4, ...

Every natural number is included in whole numbers, but whole numbers also include 0, so whole numbers are larger.

$$\text{Sol.110.(a)} \sqrt{1849} = 43$$

Sol.111.(c) Integer: A whole number that can be positive, negative, zero, without any fractional or decimal parts.

$$\frac{(2^3 - 2)}{2} = \frac{8-2}{2} = \frac{6}{2} = 3 \text{ (integer)}$$

$$\sqrt{121} = 11 \text{ (integer)}$$

$$\frac{(7^2 - 1)}{2} = \frac{49-1}{2} = 24 \text{ (integer)}$$

$$\frac{3}{\sqrt{3}} = \sqrt{3} = 1.732 \text{ (not integer)}$$

Sol.112.(a)

Even natural numbers: 2, 4, 6, 8, 10,

Positive multiples of 2: 2, 4, 6, 8, 10,

Both sets contain exactly the same numbers. Therefore, this pair represents a set of equal numbers.

$$\text{Sol.113.(d) Given:- } 2\frac{3}{7} + 5\frac{4}{9}$$

$$2\frac{3}{7} = \frac{2 \times 7 + 3}{7} = \frac{17}{7}$$

$$5\frac{4}{9} = \frac{5 \times 9 + 4}{9} = \frac{49}{9}$$

$$\frac{17}{7} + \frac{49}{9} = \frac{496}{63} = 7\frac{55}{63}$$

Sol.114.(a) Total length of wire

$$= 18\frac{2}{5} = \frac{92}{5} = 18.4 \text{ m}$$

Length of each piece = 1.6 m

Number of complete pieces

$$= \frac{18.4}{1.6} = 11.5 \approx 11$$

Length used = $11 \times 1.6 = 17.6 \text{ m}$

Unused length = $18.4 - 17.6 = 0.8 \text{ m}$

Hence, 11 complete pieces, with 0.8 m of wire left unused.

$$\text{Sol.115.(a)} \sqrt{73441} = 271$$

$$\text{Sol.116.(b) Sum} = \sqrt{5} + (3 - \sqrt{5}) = 3$$

Since 3 is a whole (and hence rational) number, the sum is a rational number.

Sol.117.(c)

Natural numbers = 1, 2, 3, 4, 5, ...

Whole numbers = 0, 1, 2, 3, ...

That is, adding 0 to natural numbers forms the set of whole numbers.

Sol.118.(c) To the right of 0 on the number line are all positive real numbers.

Sol.119.(d) No number can be the same in both sets, because every integer is rational and an irrational number can never be rational.

Sol.120.(d) A square ends in 9 only when the number ends in 3 or 7.

Ends with 3 :- 13, 23, 33, 43, 53, 63, 73, 83, 93. → 9 number

Ends with 7 :- 17, 27, 37, 47, 57, 67, 77, 87, 97. → 9 number

Total 18 numbers

Sol.121.(a) If a number is irrational, then its number will not be rational.

Sol.122.(b) The real numbers are the largest group that includes both.

$$\text{Sol.123.(a)} (18)^2 = 324$$

Sol.124.(d)

$256 = 16^2$, which is a perfect square of an even number.

Sol.125.(b)

Difference:- $89032 - 84179 = 4853$

Prime factorization of 4853 : 23×211

Among these, only 211 is a three-digit number.

So, divisor = 211

Sum of digits: $2 + 1 + 1 = 4$

Sol.126.(a)

Whole numbers (W) = {0, 1, 2, 3, 4, ...}

Natural numbers (N) = {1, 2, 3, 4, ...}

Clearly,

N subset W

and it is a proper subset because W contains one extra element (0) that natural numbers do not include.

Sol.127.(a) A-1, B-2, C-3, D-4

$$\text{Sol.128.(b) Given, } x = 7 \text{ and } y = 5 - \sqrt{7}$$

$$x + y = 7 + 5 - \sqrt{7} = 5$$

Sol.129.(a) Both A and R are true, and R is the correct explanation of A

Sol.130.(d) 2118 is not a perfect square number.

$$\text{Sol.131.(b) Prime factorization of } 180 = 2^2 \times 3^2 \times 5^1$$

Since 5 does not form a pair,

Therefore, to make the number a perfect square, we must multiply it by 5.

$$180 \times 5 = 900$$

Sol.132.(d) From option (d)

$$\sqrt{5} + (-\sqrt{5}) = 0$$

Both numbers are irrational, but their sum is 0 (a rational number).

Sol.133.(b) -3 is an integer but not natural.

Sol.134.(a)

Check each option for the condition:

Number - Its square root = 225, and the number is a multiple of 15.

Option (a) 225

$$225 - \sqrt{225} = 225 - 15 = 210 = \text{RHS}$$

Sol.135 (a): Both A and R are true, and R is the correct explanation of A

Sol.136.(d) All integers (z) are rational numbers (x), and all rational numbers are real numbers (y).

Hence, $z \subseteq x \subseteq y$.

Sol.137.(b) According to question,

$$\text{Length of each part} = \frac{\sqrt{392}}{14} = \frac{14\sqrt{2}}{14}$$

$$= \sqrt{2} \text{ m}$$

$\sqrt{2}$ is an irrational number

Sol.138.(c)

Natural number (p) = 1, 2, 3, 4,

$$\Rightarrow p(p+1)(p+2)(p+3) + 1$$

On putting p = 1,

$$\Rightarrow 1 \times 2 \times 3 \times 4 + 1 = 25$$

25 is a perfect square number

On putting p = 2

$$\Rightarrow 2 \times 3 \times 4 \times 5 + 1 = 121$$

121 is also a perfect square,

It is clear that the given expression is a perfect square.

Sol.139.(b) According to question,

$$(100010)_2 = 1 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 32 + 2 = 34$$

Sol.140.(d) Like surds must have the same irrational surd part (same radicand after simplification).

$-2\sqrt{15}$, $-3\sqrt{7}$, and $5\sqrt{37}$ all have irrational square roots in their simplest form.

$\sqrt{25}$ which is a rational number, so it is not a surd.

$$\text{Sol.141.(d)} \sqrt{80} = \sqrt{(16 \times 5)} = 4\sqrt{5} = 4 \times 2.236 = 8.944$$

$$\sqrt{800} = \sqrt{(100 \times 8)} = 10\sqrt{8} = 10 \times 2\sqrt{2} = 10 \times 2.828 = 28.28$$

So the natural numbers are: 9, 10, 11,, 27, 28

$$\text{Number of natural numbers} = 28 - 9 + 1 = 20.$$

Sol.142.(a)

If a perfect square contains 4n digits, then: $4n = 2 \times (2n)$

[Note: A perfect square with 2k digits has a square root with k digits.]

Here,

$$2k = 4n \Rightarrow k = 2n$$



So the number of digits in the square root is: $2n$

Sol.143.(d)

Convert 10110101 to decimal:

$$(1 \times 2^7) + (0 \times 2^6) + (1 \times 2^5) + (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) \\ = 128 + 0 + 32 + 16 + 0 + 4 + 0 + 1 = 181$$

Convert 11001010 to decimal:

$$(1 \times 2^7) + (1 \times 2^6) + (0 \times 2^5) + (0 \times 2^4) + (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (0 \times 2^0) \\ = 128 + 64 + 0 + 0 + 8 + 0 + 2 + 0 = 202$$

Add the decimal numbers:

$$\Rightarrow 181 + 202 = 383$$

Sol.144.(b) According to the question

$$\sqrt{-49} = \sqrt{49} \cdot \sqrt{-1} = 7i$$

It is not a real number because the square root of a negative number is imaginary, but it is a complex number.

Sol.145.(a) The square root is rational when the number is a perfect square.

$$\sqrt{144} = 12$$

Sol.146.(c) Given numbers, $\sqrt{2}, \sqrt[4]{3}, \sqrt[3]{7}$

$$\Rightarrow 2^{\frac{1}{2}}, 3^{\frac{1}{4}}, 7^{\frac{1}{3}} \text{ LCM of } \frac{1}{2}, \frac{1}{4}, \frac{1}{3} = 12$$

$$\Rightarrow 2^6, 3^3, 7^4$$

Hence, ascending order of the numbers are, $\sqrt[4]{3} < \sqrt{2} < \sqrt[3]{7}$

Sol.147.(a) 225 is the square of 15 and is also a multiple of 5**Sol.148.(b)** -15 is integer**Sol.149.(a)**

1111111(binary) converted to decimal:

$$(1111111)_2 = (127)_{10}$$

11(binary) converted to decimal:

$$11_2 = 3_{10} \Rightarrow \frac{127}{3} = 42.33$$

Sol.150.(d) Given: 11^{203}

$$\Rightarrow \frac{203}{4} = 3 \text{ (Remainder)}$$

then, $11^3 = 1331$

So, the unit digit = 1

Sol.151.(c) When two irrational numbers are added

$$\text{Case 1: } \sqrt{2} + (-\sqrt{2}) = 0 \text{ (rational)}$$

$$\text{Case 2: } \sqrt{2} + \sqrt{3} = \text{(irrational)}$$

$\therefore$ Number can be sometimes rational or sometimes irrational.

Sol.152.(a) Smallest number divisible by 8 = 104 and largest 3 digit number divisible by 8 = 992

$$\text{Number of terms} = \frac{992-104}{8} + 1 = 112$$

$$\text{Sum of numbers} = S_n = \frac{n}{2}(a + l)$$

$$S_n = \frac{112}{2}(104 + 992)$$

$$= 56 \times 1096 = 61376$$

Sol.153.(d)

Largest four digit number = 9999

$$\text{Now, } 9999 = 129 \times 77 + 66$$

If we subtract 66 then, resultant value is divisible by 77

$\therefore$ 4-digit number exactly divisible by 77

$$= 129 \times 77 = 9933$$

Sol.154.(d)

Let one part = x, Second part = 37 - x

According to question,

$$\Rightarrow 5x + 11(37 - x) = 227$$

$$\Rightarrow 5x + 407 - 11x = 227$$

$$\Rightarrow -6x = -180 \Rightarrow x = 30$$

$\therefore$ one part = 30, second part = 7

Hence, 9 times of smaller number = $9 \times 7 = 63$

Sol.155.(c) To solve this type of question, one is taken.

Now,

The total number of legs of all the horses on the farm = $4 \times 60 = 240$

Given:- number of pigeon legs = 2 and

number of horse legs = 4

According to question,

$$(4 - 2) = 2 \text{ unit} = 240 - 150$$

$$1 \text{ unit} = \frac{90}{2} = 45$$

Hence, number of horses in the farm

$$= 60 - 45 = 15$$

Sol.156.(b) According to question,

$$\text{Length} \propto \frac{1}{\text{Frequency}} \Rightarrow L = \frac{K}{F}$$

$$\Rightarrow 14 = \frac{K}{450} \Rightarrow K = 6300$$

$$\text{Now, } \Rightarrow 12 = \frac{6300}{F}$$

$$\Rightarrow F = \frac{6300}{12} = 525 \text{ cycles per second}$$

Sol.157.(c) In this type question solving by the help of options,

On checking all options one by one, only option (c) satisfies.

$$\Rightarrow b = 8a + \frac{225}{a^2}$$

$$\text{LHS} = b = 49$$

$$\text{RHS} = 8a + \frac{225}{a^2}$$

$$= 8 \times 5 + \frac{225}{25} = 40 + 9 = 49$$

Hence, LHS = RHS

Sol.158.(b) According to question,

$$\Rightarrow 6 \text{ unit} = 21$$

$$\Rightarrow 7 \text{ unit} = \frac{21}{6} \times 7 = 24.5$$

Sol.159.(a) If a, b and c are in continue proportion then it can be write as

$$a : b :: b : c$$

$$\Rightarrow b^2 = ac$$

Let least possible number be P and subtracted from each other

$$\Rightarrow (9 - P)^2 = (7 - P)(12 - P) \Rightarrow P = 3$$

$$\therefore P^2 + 1 = 9 + 1 = 10$$

Sol.160.(c) A number is divisible by 6 if it is divisible by both 2 and 3:

Divisibility of 2 : A number is divisible by 2 if it is an even number,

Divisibility of 3 : A number is divisible by 3 if the sum of all digits of that number is divisible by 3.

Now, 217924k

$$\Rightarrow 2 + 1 + 7 + 9 + 2 + 4 + k = 25 + k$$

The possible value of k can be 0, 2, 4, 6, 8

But the sum of number must be multiple of 3 so the value of k will be 2

Sol.161.(a) Divisibility of 9:- sum of digits of number either be 0 or divisible by 9.

Given number = 28735429

$$2 + 8 + 7 + 3 + 5 + 4 + 2 + 9 = 40$$

For making given number divisible by 9 we get remainder 4

$$\text{i.e. } 36 + 4 = 40$$

Sol.162.(b) Divisibility of 11:- difference of the sum of odd and even places digits be 0 or divisible by 11

Given number :- 7845K854

$$8 + 5 + 8 + 4 - (7 + 4 + K + 5) = 0 \text{ or divisible by 11}$$

$$25 - (16 + K) = 0 \text{ or divisible by 11}$$

$$9 - K = 0 \text{ or divisible by 11}$$

$$K = 9$$

Sol.163.(c)

Largest 5-digit number = 99999

On dividing 99999 by 47, we get 30 as a remainder.

So, the required no which is divisible by 47 = 99999 - 30 = 99969

Sol.164.(b) According to the question,

$$\Rightarrow 10x + 20 \times 6 = 180 \Rightarrow 10x = 180 - 120$$

$$\Rightarrow x = \frac{60}{10} = 6$$

Sol.165.(a)

$$\frac{(77^{77} + 77)}{78} = \frac{[(78 - 1)^{77} + 77]}{78}$$

$$= \frac{78^{77} + (-1)^{77}}{78} + \frac{77}{78}$$

$$\text{Remainder} = \frac{77 - 1}{78} = 76$$

Sol.166.(b) Divisibility rule of 8: last three digits should be divisible by 8.

Divisibility rule of 5: last digit is either 0 or 5.

If last digit is 5:

The 62684*5 is not divisible by 8.



If last digit is 0:

62684*0

The value of * can be 0, 4, 8.

By checking options, option (b) satisfied the given condition.

Sol.167.(c) Remainder $(\frac{999}{99}) = 9$

Hence, required smallest number
 $= 99 - 9 = 90$

Sol.168.(a) According to the question,
 Remainder = 32

Divisor = 4 × quotient

Divisor = 2 × remainder = 2 × 32 = 64

So, divisor = 64, quotient = 16

$\Rightarrow$ dividend = divisor × quotient + remainder

$\Rightarrow$ dividend = 64 × 16 + 32 = 1056

Sol.169.(a) $10 = 2 \times 5$

Divisibility rule of 10 : number is divisible by 10, when its unit place is 0

From the given option, only option (a) is divisible by 10.

Sol.170.(c) Even prime number = 2

Mean proportional of $\frac{36}{5}$ & 245

$$= \sqrt{\frac{36}{5} \times 245} = 42$$

Then, the quotient when 42 is divisible by 2 = 21

Sol.171.(c)

Let the initial price of apples be ₹x

According to question

$$\frac{720}{x-2} - \frac{720}{x} = 4$$

$$720\left(\frac{2}{x(x-2)}\right) = 4$$

$$x(x-2) = 360$$

$$x^2 - 2x - 360 = 0$$

$$x^2 - 20x + 18x - 360 = 0$$

$$x(x-20) + 18(x-20) = 0$$

$$(x-20)(x+18) = 0 \Rightarrow x = 20$$

the no of apples originally bought

$$= \frac{720}{20} = 36$$

Sol.172.(c) $6 = 2 \times 3$

Using the given options, we have ;

(a) 7234564, which is not divisible by 3.

(b) 7233563, which is not divisible by 2

(c) 7232562, which is divisible by 2 and 3 both

(d) 7231561, which is not divisible by 2

So, the correct option is (c)

Sol.173.(b)

After checking options one by one we get,

After 4 years,

Daisy plants = 6 + 3 + 3 + 3 + 3 = 18

And Jasmine plants = 26 - 2 - 2 - 2 - 2 = 18

Hence, after 4 years daisy plants and jasmine plants become equal.

Sol.174.(c) LCM of (2,5,10 and 11) = 110
 After checking all the options one by one, only option (c) satisfies.

$$\Rightarrow \frac{589270}{110} = \text{Rem.}(0)$$

So, 589270 is divisible by 2, 5, 10 and 11

Sol.175.(a) Divisibility rule of 8 : Last three digit should be divisible by 8

Now given, 91876a2

$\therefore \frac{6a2}{8}$, by checking option one by one,

option (a) satisfy this condition

$$\frac{632}{8} = 79$$

Hence, smallest value of a = 3

Sol.176.(b) A number is divisible by 6 if it is divisible by both 2 and 3:

Divisibility of 2 : A number is divisible by 2 if it is an even number,

Divisibility of 3 : A number is divisible by 3 if the sum of all digits of that number is divisible by 3.

Now, 321y72

$$\Rightarrow 3 + 2 + 1 + y + 7 + 2 = 15 + y$$

The least value of y is 0

Sol.177.(b)

The cost of drinks for 7 friends = ₹240

A regular cup of coffee cost = ₹40

A regular cup of black tea cost = ₹30

Assume that all friends drink a coffee,

So the price of 7 cup of coffee

$$= 7 \times 40 = ₹280$$

Then, Cups of tea she purchase

$$= \frac{280 - 240}{40 - 30} = \frac{40}{10} = 4$$

Sol.178.(c) $(17^{13} - 21) \div 18$

Concept : Rem $\left\{\frac{(a-1)^n}{a}\right\} \rightarrow (a-1)$ or -1 ;

when n is odd

$$\frac{(18-1)^{13}}{18} - \frac{21}{18} \Rightarrow 17 - 3 = 14(\text{rem})$$

Sol.179.(a) Divisibility rule of 9 : Sum of the digit of the number should be divisible by 9

Given, number = 57x716

Sum of digit : 5 + 7 + x + 7 + 1 + 6 = 26 + x

Put the value of x = 1 so sum of digit is divisible by 9 $\therefore x = 1$

Sol.180.(a) A number is divisible by 11 if the difference between the sum of the digits at odd and even positions is 0 or multiple of 11.

So, 6954P

$$\Rightarrow (6 + 5 + p) - (9 + 4) = 0 \text{ or multiple of } 11$$

$$\Rightarrow 11 + p = 13 \Rightarrow p = 2$$

Sol.181.(d)

Given: $f(m) = m^5 + 5m^4 - 3m + 7$

According to the question,

$$m - 2 = 0 \Rightarrow m = 2$$

$$\text{Remainder} = 2^5 + 5 \times 2^4 - 3 \times 2 + 7 = 32 + 80 - 6 + 7 = 113$$

Sol.182.(a) Number (r) $\div 8 \rightarrow 3$ (rem)

$$\text{Number } (r^2 + 6r + 7) \div 8$$

$$\frac{3^2 + 6(3) + 7}{8} = \frac{34}{8} \rightarrow 2 \text{ (rem)}$$

Sol.183.(c) **Given:** 19^{200}

$$= \frac{19^{200}}{20} \Rightarrow \text{Remainder} = (-1)^{200} = 1$$

$$\text{Sol.184.(d)} \quad \frac{2}{3} = 0.66, \frac{6}{53} = 0.11,$$

$$\frac{79}{90} = 0.87, \frac{33}{44} = 0.75$$

It is clear that, $\frac{79}{90}$ is the largest fraction.

Sol.185.(d) Number = 5x79856y6

For divisibility of 36 a number must be divisible by 9 and 4.

For divisibility of 4 :- last 2 digit must be divisible by 4.

Hence, possible value of y = 1, 3, 5, 7, 9

We need largest possible value of y = 9

For divisibility of 9 :- sum of number must be divisible by 9.

$$\text{Hence, sum of number} = 5 + x + 7 + 9 + 8 + 5 + 6 + y + 6 = 46 + x + y$$

On putting largest value of y = 9

$$= 46 + x + 9 = 55 + x$$

Now, Number next to 55 that is divisible by 9 is 63

Hence, Required value of x = 63 - 55 = 8

Now,

$$\sqrt{(2x + y)} = \sqrt{(2 \times 8 + 9)} = \pm 5$$

Hence negative value of $\sqrt{(2x + y)} = -5$

Sol.186.(b) Let Anna and Ben has x and y marbles respectively.

According to question,

$$x - 1 = y + 1 \Rightarrow x - y = 2$$

$$x = 2 + y \dots (i)$$

and

$$x + 1 = 3(y - 1)$$

$$2 + y + 1 = 3y - 3$$

$$2y = 6 \Rightarrow y = 3$$

Put y = 3 in equation (i)

$$x = 2 + 3 = 5$$

Hence,

Total marbles both have = 5 + 3 = 8

Sol.187.(c)



$$\begin{array}{r}
 x+2 \overline{) x^3 - 4x^2 - 8x + 11} \quad x^2 - 6x + 4 \\
 \underline{x^3 + 2x^2} \\
 -6x^2 - 8x \\
 \underline{-6x^2 - 12x} \\
 4x + 11 \\
 \underline{4x + 8} \\
 3
 \end{array}$$

When $x^3 - 4x^2 - 8x + 11 \div (x + 2)$, we get 3 as remainder, so if 3 is subtracted from $x^3 - 4x^2 - 8x + 11$ then it is divisible by $(x + 2)$

Sol.188.(b)

Let number of marbles Ravi had = x
 Number of Marbles Kavita had = $x + 12$
 According to question ,
 $\Rightarrow 3(x + 12) = 4(x + 1)$
 $\Rightarrow 3x + 36 = 4x + 4 \Rightarrow x = 32$
 $\therefore$ Number of Marbles Kavita had $(x + 12)$
 $= 32 + 12 = 44$

Sol.189.(b) Divisibility rule of 11 :
 Difference between sum of digit odd places and sum of digits at even places is either 0 or multiple of 11
 By checking all options one by one option (b) does not satisfy this condition
 $756148 = (7 + 6 + 4) - (5 + 1 + 8)$
 $= 17 - 14 = 3 \neq 0$, Multiple of 11
 $\therefore$ Option (b) is not divisible by 11

Sol.190.(c) Divisibility of 8 : A number is divisible by 8 if the last three digits of the number are either zeros or are divisible by 8
 So, 7344, 5544 and 4608 are divisible by 8 except for **7497**.

Sol.191.(b) Given: $2^{2n} + 2n$

Put $n = 3$,
 $\Rightarrow 2^{2 \times 3} + 2 \times 3 = 64 + 6$
 $= 70$ (not divisible by 6)
 Put $n = 1$
 $\Rightarrow 2^{2 \times 1} + 2 \times 1 = 4 + 2 = 6$
 (divisible by 6)
 Put $n = 0$
 $2^{2 \times 0} + 2 \times 0 = 1 + 0 = 1$
 (not divisible by 6)
 Put $n = 2$
 $2^{2 \times 2} + 2 \times 2 = 16 + 4 = 20$
 (not divisible by 6)
 It is clear that the correct answer is option (b)

Sol.192.(b) Divisible of 11 : if the difference between the sum of the digits at odd places and the sum of the digits at even places is 0 or multiple of 11
 7278745
 $\Rightarrow (7 + 7 + 7 + 5) - (2 + 8 + 4)$
 $\Rightarrow 26 - 14 = 12$

It is clear that if 1 is subtracted from this number (786452), then this number will become divisible by 11.

Sol.193.(d)

Let the price of one chairs and one tables be x and y respectively,
 $\Rightarrow 10x + 6y = ₹5140$ (i)
 $\Rightarrow 3x + 2y = ₹1635$ (ii)
 On multiply by 3 in equation (ii) then subtract equation (i) from (ii)
 $\Rightarrow x = 5140 - 4905 = ₹235$
 Put the value of x in equation (ii)
 $\Rightarrow 3 \times 235 + 2y = ₹1635$
 $\Rightarrow 2y = 1635 - 705$
 $\Rightarrow y = \frac{930}{2} = ₹465$
 Now, The value of 1 chair and 1 table
 $= 235 + 465 = ₹700$

Sol.194.(d)

Option (a) 3 and 4 (not correct)
 429714 is divisible by 3 but not divisible by 4
 Option (b) 6 and 5 (not correct)
 429714 is divisible by 6 but not divisible by 5
 Option (c) 4 and 6 (not correct)
 429714 is divisible by 6 but not divisible by 4
 Option (d) 3 and 6 (correct)
 429714 is divisible by both numbers 3 and 6

Sol.195.(c) Divisibility rule of 3 : Sum of the digit of the given number should be divisible by 3
 Here, $6 + 7 + 4 + 5 + k + 2 = 24 + k$
 If $k = 9$, then $24 + 9 = 33$ (which is divisible by 3)
 $\therefore$ Greatest value of $k = 9$

Sol.196.(d)

Number $\div 55 \rightarrow 28$ (remainder)
 Number $\div 11 \rightarrow \frac{28}{11} = 6$ (remainder)

Sol.197.(c)

Two numbers = 2 unit and 5 unit
 Difference = $5 - 2 = 3$ unit
 3 unit = 210
 Smaller number (2 unit) = $\frac{210}{3} \times 2 = 140$

Sol.198.(d) For divisibility of 3 the sum of digits must be divisible by 3.
 Hence, Sum of digits of number
 $= 4 + 3 + 6 + P + 5 = 18 + p$
 Value of P can be 3, 6, 9 but for largest value $P = 9$

Sol.199.(a)

Sum of first n natural number = $\frac{n(n+1)}{2}$
 Sum of first 12 natural number

$$= \frac{12(12+1)}{2} = 6 \times 13$$

Hence, 6 can divide the sum of first 12 natural numbers.

Sol.200.(a) $8^m - 1$

Put $m = 2$ (smallest even number)
 $8^2 - 1 \Rightarrow 64 - 1 = 63$
 Hence, $8^m - 1$ is divisible by 63 when m is even.

Sol.201.(c) For divisibility of 6 a number must be divisible by 2 and 3.
 Among the options only option (c) is divisible by 3 and 2 both.

Sol.202.(a)

Let the number is $15x$ (LCM of 5 and 3)
 According to question
 $\frac{1}{5}$ of $15x + 5 = \frac{1}{3}$ of $15x - 7$
 $3x + 5 = 5x - 7 \Rightarrow 2x = 12 \Rightarrow x = 6$
 Hence, number = $15x = 15 \times 6 = 90$
 Half of number = 45

Sol.203.(c)

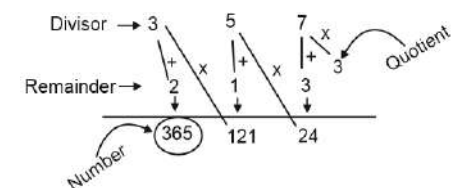
Let two consecutive even no. = $x, x + 2$
 According to question,
 $x + x + 2 = 174 \Rightarrow 2x = 172 \Rightarrow x = 86$

Sol.204.(a) $7^{15} + 7^{16} + 7^{17}$

$7^{15}(1 + 7 + 7^2)$
 $7^{15}(1 + 7 + 49) \Rightarrow 7^{15}(57)$
 Among the options 57 is divisible by 3 only.

Sol.205.(b) In this type of questions, we start the calculation from the last and come towards the initial position.
 The number which when divided by 7 leaves remainder 3 and quotient is 3
 $= 7 \times 3 + 3 = 24$

When a number is divided by 5, the remainder is 1 = $24 \times 5 + 1 = 121$
 When a number is divided by 3, the remainder is 2 = $121 \times 3 + 2 = 365$
 Hence, required number = 365

Short trick :-

Sol.206.(c) Numbers that are divisible by 8 between 2000 and 2020 is 2008, 2016 (2 numbers)

Sol.207.(b)

Let number = $(304 + 43) = 347$
 Now, required remainder = $\frac{347}{16} = 11$

Sol.208.(c)

Let five consecutive number be $x - 4, x - 2, x, x + 2, x + 4$ respectively
 Their sum = 5x
 According to the question,
 $5x = 2720 \Rightarrow x = 544$
 Required sum = $(x + x + 4)$
 $= 2 \times 544 + 4 = 1092$

Sol.209.(d) According to the question,
 Number of digits required
 $= (9 \times 1) + (90 \times 2) + (329 \times 3) = 1176$

Sol.210.(a) Divisibility rule of 9 :- sum of digits is divisible by 9
 When any 8 digit number is divisible by 99 it means that number is also divisible by 9 and 11.
 Even if we shuffle the digits the number will be divisible by 9 but not by 11

Sol.211.(d)

Required remainder = $\frac{67}{32} = 3$

Sol.212.(c) $\frac{179 \times 172 \times 173}{17}$
 $\frac{9 \times 2 \times 3}{17} = \frac{18 \times 3}{17} = \frac{1 \times 3}{17} = 3$

Hence, 3 will be the remainder.

Sol.213.(b) Given number 489y5z6 is divisible by 72 (8×9)

Divisibility of 9 :- sum of the digit of number be divisible by 9
 $\Rightarrow 489y5z6 = 4 + 8 + 9 + y + 5 + z + 6$
 $= y + z + 32$
 So, possible values of $(y + z) = 4, (4 + 9 = 13), (13 + 9 = 22)$ etc.

Now, **divisibility of 8** :- last three digit of the number be divisible by 8

Last three digit = 5z6

Possible value of z = 3, 7

For highest possible product of y and z

We take, y = 6 and z = 7

$\{y + z = 6 + 7 = 13\}$

Hence, required product of $(y \times z)$

$= 7 \times 6 = 42$

Sol.214.(a) Dividend = divisor $\times$ quotient + remainder

When, n is divided by 18, the quotient is x and the remainder is 6
 $\Rightarrow n = 18x + 6$ ----- (i)

When, n is divided by 25, the quotient is y and the remainder is 15
 $\Rightarrow n = 25y + 15$ ----- (ii)

Equate equation (i) and (ii) we get;

$18x + 6 = 25y + 15 \Rightarrow 18x - 25y = 9$

Sol.215.(b) Checking option one by one the correct option is (b)

Divisibility rule of 11 - difference of sum of odd and even places sum is equal to 0 or multiple of 11.

Now, 893002 $\rightarrow$ Even places' digit

$= 9 + 0 + 2 = 11$

Odd places' digit = $8 + 3 + 0 = 11$

$11 - 11 = 0$

So the number 893002 is completely divisible by 11.

Sol.216.(d) $\frac{7658}{45} = 8$ (remainder)

So the nearest number of 7658 which is exactly divisible by 8 will be $7658 - 8 = 7650$

Sol.217.(b) To solve these kinds of questions we take help of the options,
 Option (a) $67 + 76 = 143$ (not satisfied)

Option (b) $34 + 43 = 77$ (satisfied)

Option (c) $12 + 21 = 33$ (not satisfied)

Option (d) $45 + 54 = 99$ (not satisfied)

So number will be 34

Sol.218.(b) Divisibility rule of 3 = sum of digits is divisible by 3

$7 + x + 7 + 6 + 2 + 4 = 26 + x$

Put $x = 1$

$26 + 1 = 27$ (which is divisible by 3)

So the smallest number will be 1

Sol.219.(c) Given: $k^{33} - k$

$= k(k^{32} - 1)$

$= k(k^{16} - 1)(k^{16} + 1)$

$= k(k^8 - 1)(k^8 + 1)(k^{16} + 1)$

$= k(k^4 - 1)(k^4 + 1)(k^8 + 1)(k^{16} + 1)$

$= k(k^2 - 1)(k^2 + 1)(k^4 + 1)(k^8 + 1)(k^{16} + 1)$

$= k(k - 1)(k + 1)(k^2 + 1)(k^4 + 1)(k^8 + 1)(k^{16} + 1)$

Put the value of $k = 3, 5, 7, \dots$

$= 3 \times 2 \times 4[(3^2 + 1)(3^4 + 1)(3^8 + 1)(3^{16} + 1)]$

$= 24 \times [(3^2 + 1)(3^4 + 1)(3^8 + 1)(3^{16} + 1)]$

From the above expression it is clear that $k^{33} - k$ will always be divisible by 24.

Sol.220.(a) LCM of 2, 3, 4 = 12

Numbers between 10 to 65 which is divisible by 2, 3, 4 = 12, 24, 36, 48, 60

Hence, the required numbers will be 5.

Sol.221.(a)

Practice time to get 70 marks = 6 hours

Practice time to get 90 marks

$= \frac{6}{70} \times 90 = 7.7$ hours (approx)

Sol.222.(c) Let number = $10x + y$

According to the question,

Number is 12 more than five times the sum of its digits

$10x + y = 5(x + y) + 12$

$5x - 4y = 12$ ----- (i)

Number formed by reversing the digits is 9 less than the original number

$10y + x = 10x + y - 9$

$x - y = 1$ ----- (ii)

Equation [(i) - 5 $\times$ (ii)] we get; $y = 7$

Put $y = 7$ in equation (ii) we get;

$x - 7 = 1 \Rightarrow x = 8$

Hence, number = $10 \times 8 + 7 = 87$

Sol.223.(c) Let $n = (5 + 3) = 8$

Then, $6n = 6 \times 8 = 48$

Hence, rem. $\frac{6n}{5} = \text{rem. } \frac{48}{5} = 3$

Sol.224.(d) $17^3 + 18^3 - 16^3 - 15^3$

$4913 + 5832 - 4096 - 3375 = 3274$

Clearly, the number is divisible by 2.

Sol.225.(a) n divides 732 and leaves 12 as a remainder

Then, number n divides $(732 - 12)$

$= 720$ completely

Now, factor of $720 = 2^4 \times 3^2 \times 5^1$

Number of factor

$= (4 + 1) \times (2 + 1) \times (1 + 1) = 30$

Since, remainder is 12 then, value of n must be greater than 12

Number of factors of 720 which are less or equal to 12

$= (1, 2, 3, 4, 5, 6, 8, 9, 10, 12)$

$= 10$ factor

So the possible number of n

$= 30 - 10 = 20$

Sol.226.(b) age of a tree depends on the number of rings in the trunk

40 rings = 50 years $\Rightarrow 1$ rings = 1.25 years

Then, 30 rings = $30 \times 1.25 = 37.5$ years

The value of tree is proportional to age of tree

50 years = Rs. 50000

$\Rightarrow 1$ years = Rs. 1000

Then, 37.5 years

$= 37.5 \times 1000 = \text{Rs. } 37500$

Hence, tree with 30 rings is 37.5 years of age and its value is Rs. 37500

Sol.227.(c) $\frac{(8^6 + 1)}{7} = \frac{(1^6 + 1)}{7} = 2$

Sol.228.(b)

Let the original number be $(10x + y)$

On reversing its digits, no will be $(10y + x)$

$y + x = 9$ ----- (i)

According to the question,

$(10y + x) - (10x + y) = 27 \Rightarrow 9y - 9x = 27$

$y - x = 3$ ----- (ii)

On solving eqn(i) and (ii) we have ;

$y = 6, x = 3$

So, the number = $10 \times 3 + 6 = 36$

Sol.229.(a) Number is divisible by 11 if and only if the difference of the sum of the digits in the even and odd positions in the number is 0 or multiple of 11

Sol.230.(c)

Rem $\left(\frac{2^{256}}{17}\right) = \text{Rem}\left(\frac{(2^4)^{64}}{17}\right)$

$= \frac{(-1)^{64}}{17} = \frac{1}{17} = 1$

Sol.231.(b) Let the larger no and smaller no be x and y respectively

$x + y = 76$ ----- (i)

$$3x - 4y = 46 \text{ ---- (ii)}$$

Solving eqn. (i) & (ii) we have ;
 $x = 50, y = 26$

Sol.232.(b) $88 = 11 \times 8$

For $6p5157q$ to be divisible by 8, its last 3 digits i.e. $57q$ must be divisible by 8.

For this q must be 6

Now, For $6p51576$ to be divisible by 11, the difference of the sum of its alternate digits i.e. $(6 + 5 + 5 + 6) - (7 + 1 + p) = 14 - p$, must be divisible by 11.

For this, P must be 3.

So, $p \times q = 3 \times 6 = 18$

Sol.233.(d) Let divisor be x .

Then, no will be $(459x + 19)$

Now, remainder $\left(\frac{459x + 19}{17}\right)$

$= \text{rem}\left(\frac{19}{17}\right) = 2$

Sol.234.(c) Let the no of unit and tens digits be y and x respectively

Then, number will be $(10x + y)$

On interchanging its digits, we get

$(10y + x)$

According to the question,

$x + y = 9 \text{ ---- (i)}$

$(10y + x) - (10x + y) = 45$

$9y - 9x = 45$

$\Rightarrow y - x = 5 \text{ ---- (ii)}$

Solving equation (i) & (ii) we have ;

$x = 2, y = 7$

So, the original number $= 10 \times 2 + 7 = 27$

Sol.235.(c) Let the larger and smaller parts be x and y respectively

According to question,

$x + y = 25 \text{ ---- (i)}$

$6x + 4y = 130 \text{ ---- (ii)}$

Solving eqn(i) & (ii) we have ;

$y = 10, x = 15$

So, the larger part $= 15$

Sol.236.(b) 611611611611

As per question check divisibility by 6 and 11

For divisibility of 6 it must be divisible by 2 and 3

611611611611 is not divisible by 2 hence it is not divisible by 6 also

now, for divisibility of 11, sum of alternate digits

$1 + 6 + 1 + 1 + 6 + 1 = 16$

$6 + 1 + 1 + 6 + 1 + 1 = 16$

Difference of sum of alternate digits

$= 16 - 16 = 0$

Hence it is divisible by 11 only.

Sol.237.(c)

$3 : 4, 3 : 5, 2 : 11, 7 : 2$

$\frac{3}{4}, \frac{3}{5}, \frac{2}{11}, \frac{7}{2}$

$0.75, 0.6, 0.18, 3.5$

Hence, correct ascending order is

$2:11 < 3:5 < 3:4 < 7:2$

Sol.238.(d) Girls - Boys $= 8 \dots$ (i)

Girls + Boys $= 18 \dots$ (ii)

On solving eq. (i) and (ii), we get

Girls $= 13$ and Boys $= 5$

Product of Girls and Boys $= 13 \times 5 = 65$

Sol.239.(b) Divisibility rule of 11 : The difference of the sum of the alternate digits of any number is either 0 or multiple of 11.

Now, after checking all the options, only option (b) gets satisfied.

$969331 \rightarrow (3 + 9 + 9) - (1 + 3 + 6)$

$= 21 - 10 = 11$, which is divisible by 11.

Sol.240.(c)

From the given option, we have ;

(a) $18 \rightarrow 10000$ is not divisible by 18

(b) $17 \rightarrow 10000$ is not divisible by 17

(c) $25 \rightarrow 10000$ is divisible by 25

(d) $14 \rightarrow 10,000$ is not divisible by 14

Clearly, we can see that only option (c) is correct.

Sol.241.(d) For divisibility by 8, the last three digits should be divisible by 8.

Here $7x4$ will be divisible by 8.

The value of x will be 0, 4, 8.

Here the smallest value is 0.

Sol.242.(c) Condition for the divisibility of 6 ; Sum of digits is divisible by 3 and last digit is even.

Sol.243.(d) $88 = 11 \times 8$

From the given option, we have ;

(a) $9848 \rightarrow 9848$ is divisible by 8 but not divisible by 11

(b) $9988 \rightarrow 9988$ is divisible by 11 but not divisible by 8

(c) $8888 \rightarrow 8888$ is divisible by 8 and 11.

(d) $9768 \rightarrow 9768$ is also divisible by 8 and 11.

Clearly, we can see that 9768 is the largest 4-digit number, which is divisible by 88.

Sol.244.(b) For $1972x471$ to be divisible by 9, the sum of its digits i.e. $1 + 9 + 7 + 2 + x + 4 + 7 + 1 = 31 + x$, should be divisible by 9.

Now, putting $x = 5$, we get 36 , which is divisible by 9

So, $x = 5$

Sol.245.(a) If a number is divisible by both 11 and 13, then it must be divisible by (11×13) .

Sol.246.(a) $72 = 9 \times 8$

For $937X728Y6$ to be divisible by 8, its last 3 digits i.e. $8Y6$ should be divisible by 8.

For this y can be 1 or 5.

For $937X728Y6$ to be divisible by 9, the sum of its digits i.e. $9 + 3 + 7 + X + 7 + 2 + 8 + Y + 6 = 42 + Y + X$ should be divisible by 9.

Putting $y = 1$, we get ; $42 + 1 + X = 43 + X \Rightarrow X = 2$

Then, $X + Y = 2 + 1 = 3$

Putting $y = 5$, we get ; $42 + 5 + X = 47 + X \Rightarrow X = 7$

Then, $X + Y = 7 + 5 = 12$

From the above option, we get one of the possible values of $X + Y = 12$.

Sol.247.(a) Let the original number be x and divisor be d

$dq + 24 = x$

Multiplying both side by 2,

$2dq + 48 = 2x$

$(2dq + 35) + 13 = 2x$

So, d is either 35 or factor of 35 but greater than 24 $\Rightarrow d = 35$

Sol.248.(b) $\text{Rem}\left(\frac{5^{55}}{4}\right) = \frac{1^{55}}{4} = 1$

Sol.249.(c) $7 \times 11 = 77$

Numbers less than 1000 which are multiples of 77 $= 77, 154, \dots, 924$ since, the given series are in AP. so, we have ;

$77 + (n - 1)77 = 924 \Rightarrow (n - 1)77 = 847$

$n - 1 = 11 \Rightarrow n = 12$

Sol.250.(d) Divisibility rule of 4 :

For any no. to be divisible by 4, its last 2 digits should be divisible by 4.

Now, after going through all the options, we get only 618703592 is divisible by 4.

Hence, option(d) is the correct one.

Sol.251.(c) Let

Divisor $= n$, quotient $= q$, remainder $= r$

$(\text{Dividend})N = n.q + r$

Given:

$n = 13, q = 6, r = 39$

$\Rightarrow n = 39 \times 6 = 234 \Rightarrow q = 18$

$\Rightarrow N = n \times q + r \Rightarrow N = 234 \times 18 + 39$

$\Rightarrow N = 4212 + 39 = 4251$

Sol.252.(c) For divisible by 36 the number must be divisible by both 9 and 4.

To check divisibility by 9 sum of all the numbers divisible by 9.

and for divisibility by 4 the last two digits should be divisible by 4.

By checking option one by one option (c) satisfies.

i.e. Option (c) 55512

$5 + 5 + 5 + 1 + 2 = 18$ (divisible by both 9 and 4)

Sol.253.(a) By checking option one by one option (a) satisfies.

Option (a) 8



For divisibility by 8, last three digits should be divided by 8.
And here 216 is divisible by 8.
Hence the given number is divisible by 8.

$$\begin{aligned}\text{Sol.254.(a)} & 3 + 3^2 + 3^3 + \dots + 3^8 \\ \Rightarrow & 3(1 + 3 + 3^2 + 3^3 + \dots + 3^7) \\ \Rightarrow & 3(1 + 3 + 9 + 27 + 81 + 243 + 729 + 2187) \\ \Rightarrow & 3(3280) = 9840\end{aligned}$$

Sol.255.(c) (a) For divisibility by 3
 $2 + 9 + 1 + 8 + 2 + 4 + 5 = 31$ (not divisible by 3)
(b) Since the number is not divisible by 3, the number is not divisible by 9.
(c) For divisibility by 11
Difference between the sum of the digits placed at odd places and even places is either divisible by 11 or 0.
For 2918245
 $(2 + 1 + 2 + 5) - (9 + 8 + 4) = 10 - 21 = -11$ (divisible by 11)
(d) For divisibility by 12, the number should be divisible by 4 and 3, but the given number is not divisible by 4 and 3, hence it is not divisible by 12.

$$\begin{aligned}\text{Sol.256.(b)} & 6^{25} + 6^{26} + 6^{27} + 6^{28} \\ = & 6^{25}(1 + 6 + 6^2 + 6^3) \\ = & 6^{25}(1 + 6 + 36 + 216) = 6^{25}(259) \\ \text{Hence the given expression is divisible by } & 259.\end{aligned}$$

$$\begin{aligned}\text{Sol.257.(a)} & \frac{90}{4} = \frac{\text{Distance}}{32} \\ \Rightarrow \text{Distance} & = 90 \times 8 = 720 \text{ km}\end{aligned}$$

Sol.258.(b) We know
Dividend (N) = divisor $\times$ quotient + remainder

Given
Divisor (n) = $10 \times$ quotient (q)
And Divisor (n) = $5 \times$ remainder (R)
 $R = 12 \Rightarrow n = 60$ and $q = 6$
Now,
 $N = 60 \times 6 + 12 = 372$

$$\begin{aligned}\text{Sol.259.(a)} & 4^{12} + 4^{13} + 4^{14} + 4^{15} \\ \Rightarrow & 4^{12}(1 + 4 + 4^2 + 4^3) \\ \Rightarrow & 4^{12}(1 + 4 + 16 + 64) = 4^{12}(85) \\ \text{85 is divisible by 17 and hence 17 is the answer.}\end{aligned}$$

Sol.260.(b) For divisibility by 3,
The sum of digits of the number should be divisible by 3.
Here $3 + 7 + 9 + 8 + 4 + 3 = 34$
And $34 \div 3$ remainder is 1.
Hence,
When 379843 is divided by 3 the remainder is 1.

Sol.261.(d) For divisibility by 4, the last two digits should be divisible by 4.
Hence the possible value of y can be 1, 3, 5, 7, 9

For divisibility by 11,
The difference between the sum of the digits at odd places and even places should be zero or divisible by 11.
 $\Rightarrow (1 + x + y) - (5 + 1 + 2) = x + y + 1 - 8$
 $\Rightarrow x + y - 7$
For minimum value of x and y
Take $y = 1$
 $\Rightarrow x = 6$ (to make it $6 + 1 - 7 = 0$)
Hence, $x + y = 6 + 1 = 7$

Sol.262.(a) $72 = 9 \times 8$
For $389x6378y$ to be divisible by 8, its last 3 digit no i.e. $78y$ should be divisible by 8.
For this, $y = 4$
Now, For $389x63784$ to be divisible by 9, the sum of its digits i.e. $3 + 8 + 9 + x + 6 + 3 + 7 + 8 + 4 = 48 + x$, should be divisible by 9.
For this, we have $x = 6$
Hence, $6x + 7y = 6 \times 6 + 7 \times 4 = 36 + 28 = 64$

Sol.263.(a) Divisibility rule of 9 :
For any no to be divisible by 9, the sum of its digits should be divisible by 9.
From the given options, we have ;
(a) $553986 \rightarrow 5 + 5 + 3 + 9 + 8 + 6 = 36$, which is divisible by 9.
(b) $350846 \rightarrow 3 + 5 + 0 + 8 + 4 + 6 = 26$, which is not divisible by 9.
(c) $941201 \rightarrow 9 + 4 + 1 + 2 + 0 + 1 = 17$, which is not divisible by 9.
(d) $132490 \rightarrow 1 + 3 + 2 + 4 + 9 + 0 = 19$, which is not divisible by 9.
Clearly, we can see that 553986 is the only number which is divisible by 9.

Sol.264.(d) $15 = 5 \times 3$
For $725yz$ to be divisible by 5, unit digit i.e. z should be 0 or 5.
For $725yz$ to be divisible by 3, the sum of digits i.e. $7 + 2 + 5 + y + z = 14 + y + z$ should be divisible by 3.
But for the maximum value of yz , we have to put $z = 5$. then we get $19 + y$
 $\Rightarrow y = 8$
Then, $yz = 8 \times 5 = 40$

Sol.265.(b) Divisibility rule of 8 ;
For any number to be divisible by 8, its last 3 digits should be divisible by 8.
For $7002x4$, $2x4$ should be divisible by 8
For this, x must be 2.

Sol.266.(b) Divisibility rule of 9,
For any number to be divisible by 9, the sum of its digits should be divisible by 9.
For $72x72$, $7 + 2 + x + 7 + 2 = 18 + x$, should be divisible by 9.
For this, the value of x must be 0 or 9

Sol.267.(a) After checking all the options one by one, only option(a) satisfied.

So, 7918378 is divisible by 13.

Sol.268.(a) Dividing 2001 by 17, we get 12 as remainder.
So, On subtracting 12 from 2001, we get 1989 which is exactly divisible by 17.

Sol.269.(d) Divisibility rule of 11 :
For any no to be divisible by 11, the difference of the sum of its alternate digits should be divisible by 11.
For $7235x$ to be divisible by 11, $(7 + 3 + x) - (2 + 5) = 3 + x$, should be divisible by 11.
For this, x must be 8.

Sol.270.(d) Average of 1 negative number = average of 24 positive numbers
{Because here maximum (+ive) integers are asked whereas the average of 25 numbers is zero.}
So the maximum positive integers will be 24

Sol.271.(a) Divisibility rule of 4,
For any no to be divisible by 4, its last 2 digit should be divisible by 4
Then, For 87893P4,
P4 should be divisible by 4
For this, P can be 2, 4, 6 or 8
But for the maximum value of P, it must be 8.

Sol.272.(d) For $48397k5$ to be divisible by 9, the sum of its digits i.e. $4 + 8 + 3 + 9 + 7 + k + 5 = 36 + k$, should be divisible by 9
For non zero value of $k \Rightarrow k = 9$

$$\begin{aligned}\text{Sol.273.(d)} & \frac{999999}{294} = \text{Rem. (105)} \\ \text{So, required number} & = 294 - 105 = 189\end{aligned}$$

Sol.274.(b)
Factor of 2520 = $2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7$
Clearly 8, 9 and 7 are the divisors of number 2520.

Sol.275.(d) Sum of digits should be divisible by 3 not by 9.
 $3 + 2 + a + 7 + 8 + b = 20 + a + b$
For greatest number, a should be maximum.
Max. possible value for $(a + b) = 13$
So, $a = 9$ and $b = 4$
Required no. = 329784

Sol.276.(a) Given, $a^3 + b^3 = 10234$
 $a + b = 34$ ----- equ. (1)
We know
 $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
 $(34)^3 = 10234 + 3ab(34)$
 $39304 - 10234 = 102 \times ab \Rightarrow ab = 285$
Then, $(a - b)^2 = (a + b)^2 - 4ab$

$$(a - b)^2 = 34^2 - 4 \times 285$$

$$(a - b)^2 = 16$$

$$(a - b) = 4 \text{ ----- equ. (2)}$$

Now, Add equation (1) and (2)

$$2a = 38$$

Where, $a = 19$ and $b = 15$, So that

$$a^3 - b^3 = 19^3 - 15^3 = 6859 - 3375 = 3484$$

Sol.277.(d) Given :- $N = 4a6b9c$

N is divisible by 99 therefore it must be divisible by both 9 and 11

$$(4 + 6 + 9) - (a + b + c) \Rightarrow 19 - (a + b + c)$$

if $a + b + c = 8$ then N is divisible by both 11 and 9

$$\text{Hence, sum of digits of } N = 19 + 8 = 27$$

$$\text{Sol.278.(a)} \frac{(265)^{4081} + 9}{266} = \text{Rem}$$

$$\frac{(266 - 1)^{4081} + 9}{266} = \text{Rem}$$

$$\frac{(-1)^{4081} + 9}{266} = \text{Rem}$$

$$\frac{(-1) + 9}{266} = \text{Rem. (8)}$$

Sol.279.(a) Let A has x no. of toffees and B has y no. of toffees.

According to the question,

Condition 1 :-

$$x - 1 = y + 1$$

$$x = y + 2 \text{e.q. (1)}$$

Condition 2 :-

$$x + 1 = 2 \times (y - 1)$$

$$x - 2y = -3$$

$$y + 2 - 2y = -3 \text{ (from eq 1)}$$

$$y = 5 \text{ and } x = 7$$

$$\text{Total no. of toffees} = 5 + 7 = 12$$

Sol.280.(c) Divisibility rule of 8 = last 3 digits of the number will be divisible by 8.

$$\text{Last 3 digit of the number} = \text{.....}326 - 6$$

$$= \text{.....} 320 \text{ (divisible of the 8)}$$

$$\text{So the required number will be} = 6$$

Sol.281.(d)

Largest 5 - digit number = 99999

According to question,

$$\frac{99999}{88} \Rightarrow \text{Remainder} = 31$$

$$\text{Required number} = 99999 - 31 = 99968$$

Sol.282.(d) Let five consecutive odd natural numbers are

$$(x - 4), (x - 2), x, (x + 2), (x + 4)$$

$$(x - 4)^2 + (x - 2)^2 + x^2 + (x + 2)^2$$

$$+ (x + 4)^2 = 233 \times 5 \Rightarrow 5x^2 + 40 = 1165$$

$$5x^2 = 1125 \Rightarrow x = 15,$$

$$\text{Largest number} = 19$$

$$\text{Smallest number} = 11$$

$$\text{Average} = 15$$

$$\text{Sol.283.(b)} \frac{(2^{24} - 1)}{7} = \frac{((7+1)^8 - 1)}{7}$$

$$= \text{remainder } (1)^8 - 1 = 0$$

Sol.284.(c) When a positive integer n is divided by 12 remainder is 5

$$n = 17 \Rightarrow 8n^2 + 7 = 2319$$

When 2319 is divided by 12, gives remainder 3

Sol.285.(b) $75 \times 73 \times 78 \times 76$ when individually is divided by 34 we get

$$7 \times 5 \times 10 \times 8 = 2800$$

2800 when divided by 34 gives the remainder as 12.

Sol.286.(c)

$$\text{Factor of } 10 = 2 \times 5, 12 = 2 \times 2 \times 3,$$

$$15 = 3 \times 5, 18 = 2 \times 3 \times 3 \text{ and } 20 = 2 \times 2 \times 5$$

$$\text{L.C.M. (10,12,15,18,20)} = 2 \times 2 \times 3 \times 3 \times 5$$

Least no. to be multiplied to make perfect square = 5

Then,

$$\text{Required least number} = 180 \times 5 = 900$$

